



PROFILE CAM DESIGN

8.1 INTRODUCTION

Cams are used for essentially the same purpose as linkages, that is, generation of irregular motion. Cams have an advantage over linkages because cams can be designed for much tighter motion specifications. In fact, in principle, any desired motion program can be exactly reproduced by a cam. Cam design is also, at least in principle, simpler than linkage design, although, in practice, it can be very laborious. Automation of cam design using interactive computing has not, at present, reached the same level of sophistication as that of linkage design.

The disadvantages of cams are manufacturing expense, poor wear resistance, and relatively poor high-speed capability. Although numerical control (NC) machining does cut the cost of cam manufacture in small lots, costs are still quite high in comparison with linkages. In large lots, molding or casting techniques cut cam costs, but not to the extent that stamping and so forth, can cut linkage costs for similar lot sizes.

Unless roller followers are used, cams wear quickly. However, roller followers are bulky and require larger cams, creating size and dynamic problems. In addition, the bearings in roller followers create their own reliability problems.

The worst problems with cams are, however, noise and follower bounce at high speeds. As a result, there is a preoccupation with dynamic optimization in cam design.

Cam design usually requires two steps (from a geometric point of view):

1. synthesis of the motion program for the follower and
2. generation of the cam profile.

If the motion program is fully specified throughout the motion cycle, as is the case, for example, with the stitch pattern cams in sewing machines, the first step is not needed. More usually, the motion program is specified only for portions of the cycle, allowing the synthesis of the remaining portions for optimal dynamic performance. An example is the cam controlling the valve opening in an automotive engine. Here the specification is that the valve should be fully closed for a specified interval and more or less fully open for another specified interval. For the portions of the cycle between those specified, a suitable program must be synthesized. This can be done, with varying levels of sophistication, to make the operation of the cam as smooth as possible. In general, the higher the level of dynamic performance required, the more difficult the synthesis process.

The second stage of the process, profile generation, is achieved by kinematic inversion. The cam is taken as the fixed link and a number of positions of the follower relative to the cam is constructed. A curve tangent to the various follower positions is drawn and becomes the cam profile. If the process is performed analytically, any level of accuracy can be achieved.

8.2 CAM-FOLLOWER SYSTEMS

A general cam-follower system consists of three elements as shown in Fig. 8.1. The first two are the cam and follower, and the third is a spring or other means of ensuring that the follower remains in contact with the cam. The function of the spring can be replaced by gravity or by constraining the follower between the two surfaces on the cam or constraining the cam between two surfaces on the follower. Both of these approaches are usually more expensive than using a spring and therefore are not commonly used.

A follower is characterized by its motion relative to the ground link and by the geometry of its face that contacts the cam. The cam-follower motion may be either rotational or translational, and translating followers may be either radial or offset. Examples of these are shown in Fig. 8.2. The follower surfaces may be either knife edged, flat, spherical (or cylindrical), or roller as shown in Fig. 8.3.

Actually, these geometries are all of the same class depending on the radius of curvature of the follower face. That is, the knife edge has a radius of curvature that is zero, the flat face has a radius of curvature that is infinite, and the general roller and cylindrical followers have a finite (but nonzero) radius of curvature. In this discussion, only planar cams will be considered, so no distinction between spherical and cylindrical follower faces will be made. Also, if only geometric information is of interest, no distinction needs to be made

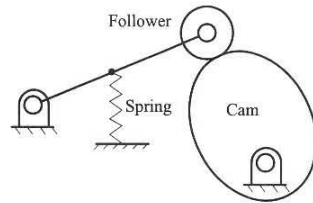


FIGURE 8.1 Elements of a cam-follower system.

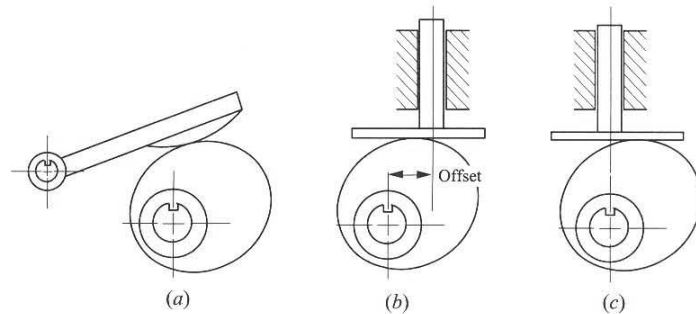


FIGURE 8.2 (a) Cylindrical-faced, oscillating follower. (b) Offset, flat-faced, translating follower. (c) Radial, flat-faced, translating follower.

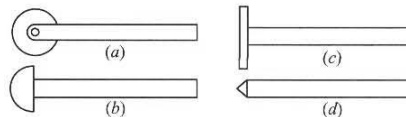


FIGURE 8.3 (a) Roller follower. (b) Cylindrical-faced follower. (c) Flat-faced follower. (d) Knife-edged follower.

between roller and rigid cylindrical-faced followers. Obviously, there is a significant difference from an overall design standpoint, however.

Although here we will consider only planar, rotating cams, in practice a large number of different cam geometries are found. Some of the different types of cams and follower systems are shown in Fig. 8.4.

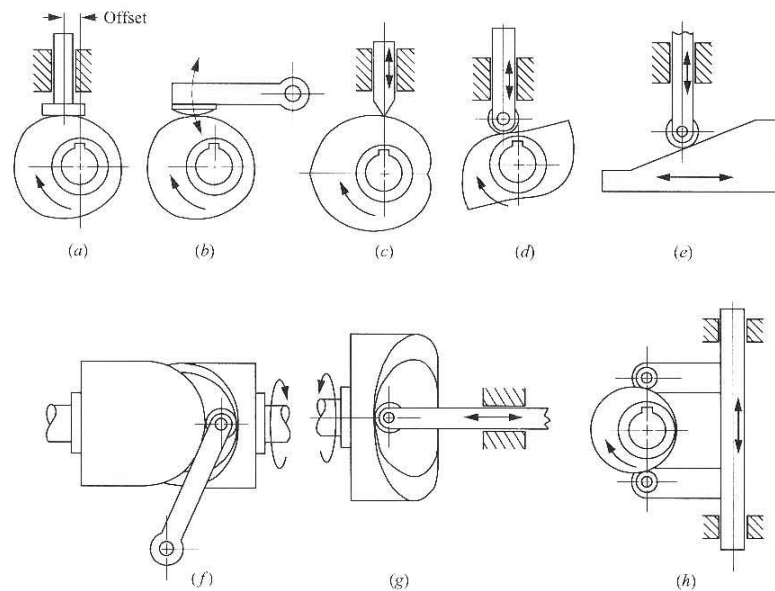


FIGURE 8.4 Some types of cams. (a) Radial cam and flat-faced, offset translating follower. (b) Radial cam and spherical-faced, oscillating follower. (c) Radial (heart) cam and translating, knife-edged follower. (d) Radial two-lobed cam and translating, offset, roller follower. (e) Wedge cam and translating roller follower. (f) Cylindrical cam and oscillating roller follower. (g) End or face cam and translating roller follower. (h) Yoke cam and translating roller follower.

8.3 SYNTHESIS OF MOTION PROGRAMS

The problem of motion-program synthesis is the problem of filling in, in an optimal way, the portions of the motion cycle that are not completely specified. The characteristics of the problem may be demonstrated by consideration of a cam that is required to drive a follower that dwells at 0 for a cam rotation of 60° , dwells at 1.0 in for a cam rotation of 110° to 150° , and is required to move with constant velocity from a displacement of 0.8 to 0.2 in for 200° to 300° of cam rotation. The specified portions of the motion program are displayed in Fig. 8.5.

A simple solution to the problem of filling the gaps is simply to move the cam at constant velocity between the specified segments, giving a follower displacement diagram as shown in Fig. 8.6.

Notice, however, that if this is done, the velocity is discontinuous at cam angles 60° , 110° , 150° , 200° , 300° , and 360° , causing the acceleration to become infinite at these locations. Since the follower cannot follow an infinite acceleration, this leads to loss of contact and/or excessive local stresses and resultant noise and wear problems.

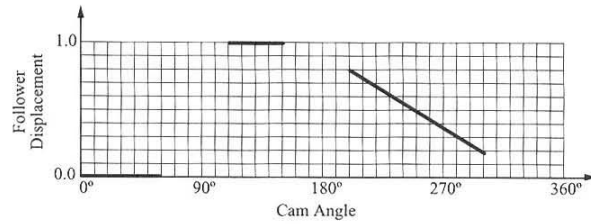


FIGURE 8.5 The statement of the required displacements of a cam design problem in graphical form.

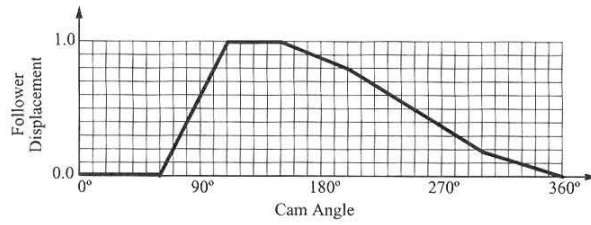


FIGURE 8.6 A cam angle-follower displacement program that satisfies the displacement requirements specified in Fig. 8.5.

The preceding motion program matches only the *displacements* at the ends of the segments. The infinite acceleration problem can be removed by matching both displacement and velocity at the ends of segments of the program. One way to do this is to subdivide the synthesized segments into two parts with a constant acceleration on the first and constant deceleration on the second. On such a subsegment, if the acceleration is a , the velocity is given by

$$v = v_0 + at$$

where v_0 is the velocity at the beginning of the segment. The displacement is given by

$$y = s_0 + v_0 t + \frac{a}{2} t^2$$

where s_0 is the displacement at the beginning of the segment. Now, if the cam is driven at constant velocity,

$$\theta = \theta_0 + \omega t$$

where θ is the cam angle, θ_0 is the cam angle at the beginning of the segment, and ω is the angular velocity. Hence

$$\begin{aligned} t &= \frac{(\theta - \theta_0)}{\omega} \\ v &= v_0 + a \frac{(\theta - \theta_0)}{\omega} \\ y &= s_0 + v_0 \frac{(\theta - \theta_0)}{\omega} + a \frac{(\theta - \theta_0)^2}{2\omega^2} \end{aligned}$$

Therefore, the relationship between s and θ , as plotted on the follower-displacement diagram, is parabolic (see Fig. 8.7). Cam-follower displacement programs that use this type of transition are called parabolic. The cam profiles developed from them are also called

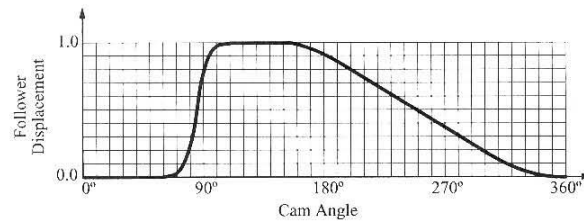


FIGURE 8.7 A follower displacement program that satisfies the displacement requirements of Fig. 8.5 using parabolic transitions. This is called a *parabolic* follower-displacement program.

“parabolic.” It is important to understand that a so-called parabolic cam *does not* have a parabolic curve in its profile. Rather, the parabolas are in the transition curves used in the follower-displacement program.

8.4 ANALYSIS OF DIFFERENT TYPES OF FOLLOWER DISPLACEMENT FUNCTIONS

Several different standard functions can be used to connect the parts of the displacement diagram where a specific type of motion is required. These displacement profiles ultimately determine the shape of the cam. Many different types of motions have been used in practice, and some have been extensively studied. These include the following:

1. uniform motion,
2. parabolic motion,
3. simple harmonic motion,
4. cycloidal motion, and
5. general polynomial motion.

The first two types of program have already been introduced. The first four types of program can be generated graphically as well as analytically, but the fifth type is generated only analytically. Both graphical and analytical development will be considered here, where possible. Both methods assume that the angular velocity, ω , of the cam is constant. If this is the case, then

$$y = y(\theta)$$

and

$$\theta = \theta_0 + \omega t$$

Here, y is used as a generic output variable. It may correspond to either a linear or angular displacement of the follower. Note that if the cam motion is given as a function of time, the motion can easily be represented as a function of the cam rotation in degrees using the preceding expressions.

The higher derivatives are given by

$$\dot{y} = \frac{dy(\theta)}{dt} = \frac{dy}{d\theta} \frac{d\theta}{dt} = y' \omega$$

and

$$\ddot{y} = \frac{d^2 y(\theta)}{dt^2} = \frac{d}{dt} \left(\frac{dy}{d\theta} \frac{d\theta}{dt} \right) = \frac{d^2 y}{d\theta^2} \left(\frac{d\theta}{dt} \right)^2 + \frac{dy}{d\theta} \frac{d^2 \theta}{dt^2} = \frac{d^2 y}{d\theta^2} \omega^2 + \frac{dy}{d\theta} \alpha$$

But because ω is constant, $\alpha = 0$ and

$$\ddot{y} = y'' \omega^2$$

Therefore, $\dot{y}$ is a simple constant times y' , and $\ddot{y}$ is also a constant times y'' . Consequently, even though we ultimately want to know the response to the time derivatives ($\dot{y}$ and $\ddot{y}$), we may work directly with the derivatives (y' and y'') with respect to the cam displacement. If the cam velocity is not a constant, then the cam profile can be designed for only one operating situation if higher derivatives are important. In the following, a constant-velocity cam is assumed, and y is again used to represent either an angular or linear displacement of the follower. Similarly, θ is used for the displacement of the cam, and it may be either an angular or a linear displacement.

The follower curves can be studied in terms of the simple diagram shown in Fig. 8.8. A general displacement diagram will be made up of three or more parts:

1. rises (1 or more),
2. returns (1 or more), and
3. dwells (0 or more).

Both the rise and return parts will contain one or more inflection points. These are points where a maximum slope is reached, and they correspond to points on the cam surface with maximum steepness. These points are identified by the locations where the curvature of the diagram changes sign. At the inflection points, the radius of curvature of the curve is infinite.

In each of the standard curve cases, we will look at mainly the rise part of the follower profile. The return part can be determined using the mirror images of the curves considered.

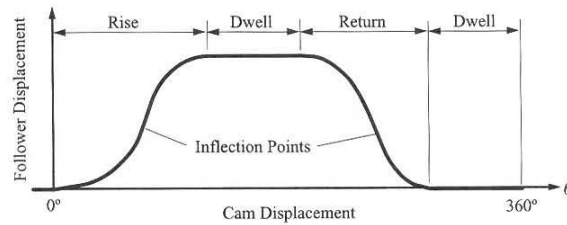


FIGURE 8.8 Terminology used when discussing follower-displacement programs.

8.5 UNIFORM MOTION

Uniform motion is represented in Fig. 8.9. To derive the equations for the follower displacement, a general form for the mathematical expression corresponding to the type of motion is assumed. The general equation will have undetermined constants in it, and these constants can be determined by matching boundary conditions at the two ends of the curve. For uniform motion, the general form of the curve used is

$$y = C\theta$$

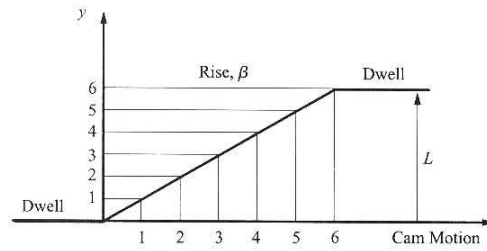


FIGURE 8.9 Uniform motion.

If L is the amount of the rise, and β is the cam rotation required for the rise, then the constant C must be L/β and y becomes

$$y = \frac{L}{\beta} \theta$$

During the rise, the velocity and acceleration are

$$\dot{y} = \frac{L}{\beta} \omega$$

and

$$\ddot{y} = 0$$

The displacement, velocity, and acceleration are plotted in Fig. 8.10. As noted earlier, the acceleration is infinite at the points where the uniform motion meets the dwells. Therefore, even for low speeds and elastic members, the forces transmitted will be very large. For *very* low speeds, however, this type of displacement diagram might be acceptable.

Graphically, the uniform motion–displacement diagram is characterized by a uniform change in y for a uniform change in the cam motion. This condition is shown in Fig. 8.9.

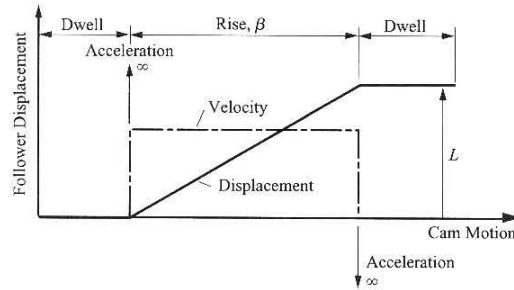


FIGURE 8.10 Displacement, velocity, and acceleration relations for uniform motion.

8.6 PARABOLIC MOTION

The equations for parabolic motion can be derived using the same procedure as described in Section 8.3. However, two parabolas must be used for each transition. The two parabolas meet at the point midway between the ends of the two dwell regions. The general form for both parabolas is

$$y = C_0 + C_1\theta + C_2\theta^2 \quad (8.1)$$

and

$$\begin{aligned}y' &= C_1 + 2C_2\theta \\y'' &= 2C_2\end{aligned}\tag{8.2}$$

If the cam displacement is taken as 0 at the beginning of the rise, then at $\theta = 0$, $y = y' = 0$. Then, $C_0 = C_1 = 0$. Also, at $\theta = \beta/2$, $y = L/2$. Therefore, the displacement and first and second derivatives with respect to θ are

$$\begin{aligned}y &= 2L\left(\frac{\theta}{\beta}\right)^2 \\y' &= 4\frac{L}{\beta^2}\theta \\y'' &= 4\frac{L}{\beta^2}\end{aligned}\tag{8.3}$$

and the velocity and acceleration are

$$\begin{aligned}\dot{y} &= 4\frac{L\omega}{\beta^2}\theta \\\ddot{y} &= 4\frac{L\omega^2}{\beta^2}\end{aligned}$$

At the point at which the curve meets the first dwell, the velocity and acceleration are continuous, but the third derivative, or jerk, is infinite. This derivative is proportional to the change in force and for high-speed cams is an important aspect of the motion. Although not so serious as having an infinite acceleration pulse, an infinite jerk can excite vibratory behavior in the system.

For the second half of the rise, the conditions to match are at $\theta = \beta/2$, $y = L/2$, and at $\theta = \beta$, $y = L$, and $y' = 0$. Then from Eqs. (8.61) and (8.62), we get

$$\begin{aligned}\frac{L}{2} &= C_0 + C_1\frac{\beta}{2} + C_2\left(\frac{\beta}{2}\right)^2 \\L &= C_0 + C_1\beta + C_2\beta^2 \\0 &= C_1 + 2C_2\beta\end{aligned}$$

The solution to this linear set of equations yields

$$\begin{aligned}C_0 &= -L \\C_1 &= \frac{4L}{\beta} \\C_2 &= -\frac{2L}{\beta^2}\end{aligned}$$

so that

$$y = L\left[1 - 2\left(1 - \frac{\theta}{\beta}\right)^2\right]\tag{8.4}$$

and

$$y' = \frac{4L}{\beta} \left(1 - \frac{\theta}{\beta} \right)$$

$$y'' = -\frac{4L}{\beta^2}$$

Finally, the velocity, acceleration, and jerk are given by

$$\dot{y} = \frac{4L\omega}{\beta} \left(1 - \frac{\theta}{\beta} \right)$$

$$\ddot{y} = -\frac{4L\omega^2}{\beta^2}$$

These equations apply to the segment of the program to the right of the inflection point shown in Fig. 8.11.

Graphically, the part of the curve up to the inflection point can be generated using the construction shown in Fig. 8.12. For the construction, the horizontal axis is divided into uniform increments, and the maximum rise is evenly divided into the same number of equal increments. The point at the origin is then connected to each of the points on the line of the maximum rise. Points on the displacement curve are given by the intersection of the diagonal lines with the corresponding vertical lines.

A cam return using parabolic motion is shown in Fig. 8.13. To determine the equations for the return from $y = L$ to 0 during the angular displacement β , we can use Eq. (8.1) again but with different boundary conditions. To simplify the equations, we will shift the origin of the coordinate system to the end of the dwell at the beginning of the return. For the first part of the return, $y = L$ and $y' = 0$ at $\theta = 0$ and $y = L/2$ at $\theta = \beta/2$. For these conditions, $C_0 = L$, $C_1 = 0$, and $C_2 = \frac{-2L}{\beta^2}$.

The displacement equation is

$$y = L \left[1 - 2 \left(\frac{\theta}{\beta} \right)^2 \right] \quad (8.5)$$

For the second half of the return, the conditions to match are at $\theta = \beta/2$, $y = L/2$, and at $\theta = \beta$, $y = 0$, and $y' = 0$. For these conditions,

$$y = 2L \left(1 - \frac{\theta}{\beta} \right)^2 \quad (8.6)$$

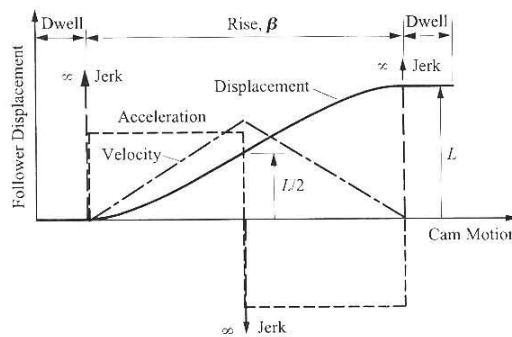


FIGURE 8.11 Displacement, velocity, acceleration, and jerk relations for parabolic motion during rise.

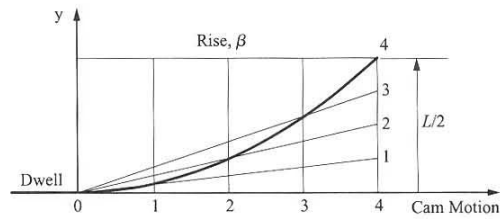


FIGURE 8.12 Construction of parabolic segment of follower-displacement program.

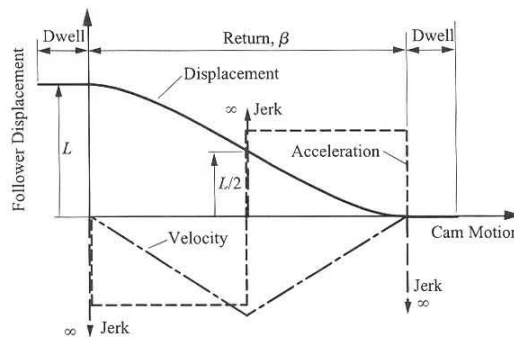


FIGURE 8.13 Displacement, velocity, acceleration, and jerk relations for parabolic motion during return.

In general, the rise and return will not always start at $\theta = 0$. However, in these cases, a simple coordinate transformation can be used. If the rise or return actually starts at $\theta = \gamma$, substitute $(\theta - \gamma)$ wherever θ appears in Eqs. (8.3)–(8.6).

EXAMPLE 8.1 Design for Parabolic Motion

Solution

Design a parabolic cam–follower displacement program to provide a dwell at zero lift for the first 120° of the motion cycle and to dwell at 0.8 in lift for cam angles from 180° to 210° . The cam profile will be laid out using 10° plotting intervals. Assume that the cam rotates with constant angular velocity.

The motion specification is as shown in Fig. 8.14. For the first part of the rise ending at $\phi = 150^\circ$ in the interval 120° to 180° , Eq. (8.3) applies if we use $\theta = (\phi - 120^\circ)$ and $0.8 = L$. The resulting expression for the first part of the rise is

$$y = 2L \left(\frac{\theta}{\beta} \right)^2 = 1.6 \left(\frac{\phi - 120^\circ}{60^\circ} \right)^2 \quad (8.7)$$

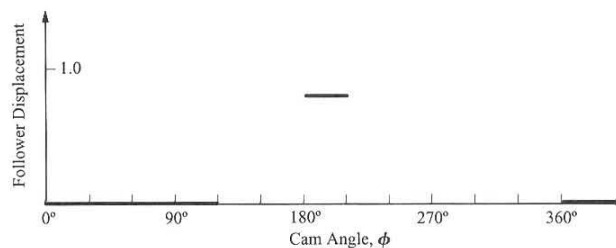


FIGURE 8.14 The motion specification for Example 8.1.

For the second part of the rise starting at $\theta = 150^\circ$, Eq. (8.4) applies if we use $\theta = (\phi - 120^\circ)$ and $0.8 = L$. The resulting expression is

$$y = L \left[1 - 2 \left(1 - \frac{\theta}{\beta} \right)^2 \right] = 0.8 \left[1 - 2 \left(1 - \frac{(\phi - 120^\circ)}{60^\circ} \right)^2 \right] \quad (8.8)$$

Using Eqs. (8.7) and (8.8) produces the successive lifts given in Table 8.1. For the first part of the return ending at $\theta = 285^\circ$ in the interval 210° to 360° , Eq. (8.5) applies if we use $\theta = (\phi - 210^\circ)$ and $0.8 = L$. The resulting expression for the first part of the return is

$$y = L \left[1 - 2 \left(\frac{\theta}{\beta} \right)^2 \right] = 0.8 \left[1 - 2 \left(\frac{(\phi - 210^\circ)}{150^\circ} \right)^2 \right] \quad (8.9)$$

For the second part of the return starting at $\theta = 285^\circ$, Eq. (8.6) applies if we use $\theta = (\phi - 210^\circ)$ and $0.8 = L$. The resulting expression is

$$y = 2L \left(1 - \frac{\theta}{\beta} \right)^2 = 1.6 \left(1 - \frac{\phi - 210^\circ}{150^\circ} \right)^2 \quad (8.10)$$

Using Eqs. (8.9) and (8.10) produces points on the return curve given in Table 8.2. The resulting transition curves are plotted in Fig. 8.15. Notice that the lift values are tabulated to four decimal places. Cam and follower systems normally use very rigid components and even small profile variations are important. For this reason, we normally work with at least four decimal places when doing cam calculations. Gears are another type of profile mechanism in which the components are very rigid and, consequently, even tiny profile variations can be important.

TABLE 8.1 Cam-Follower Data for Rise in Example 8.1

θ	120°	130°	140°	150°	160°	170°	180°
y	0.0000	0.0444	0.1778	0.4000	0.6222	0.7556	0.8000

TABLE 8.2 Cam-Follower Data for Return in Example 8.1

θ	210°	220°	230°	240°	250°	260°	270°	280°
y	0.8000	0.7929	0.7716	0.7360	0.6862	0.6222	0.5440	0.4516
θ	290°	300°	310°	320°	330°	340°	350°	360°
y	0.3484	0.2560	0.1778	0.1138	0.0640	0.0284	0.0071	0.0000

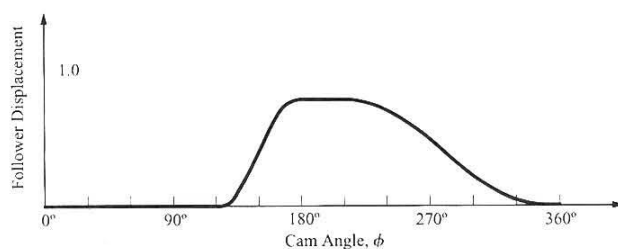


FIGURE 8.15 The parabolic follower-displacement program generated in Example 8.1.

8.7 HARMONIC FOLLOWER-DISPLACEMENT PROGRAMS

Simple harmonic motion can be generated by an offset (eccentric) circular cam with a radial follower and is therefore a common form to use for a displacement diagram. Cams with this type of transition curve are commonly referred to as “harmonic cams.” The equations for simple harmonic motion are formed from the basic equation

$$y = C_0 + C_1 \cos C_2 \theta = C_0 \left(1 + \frac{C_1}{C_0} \cos C_2 \theta \right)$$

The displacement, velocity, acceleration, and jerk diagrams are shown in Fig. 8.16. Simple harmonic motion produces a sine velocity curve and a cosine acceleration curve. There is no discontinuity at the transition point, so that θ is defined for all angles between zero and β . The equations for a rise starting from $\theta = 0$ and ending at $\theta = \beta$ and $y = L$ are

$$\begin{aligned} y &= \frac{L}{2} \left(1 - \cos \frac{\pi \theta}{\beta} \right) \\ y' &= \frac{\pi L}{2\beta} \sin \frac{\pi \theta}{\beta}, \quad \dot{y} = \frac{\pi L \omega}{2\beta} \sin \frac{\pi \theta}{\beta} \\ y'' &= \frac{L}{2} \left(\frac{\pi}{\beta} \right)^2 \cos \frac{\pi \theta}{\beta}, \quad \ddot{y} = \frac{L}{2} \left(\frac{\pi \omega}{\beta} \right)^2 \cos \frac{\pi \theta}{\beta} \\ y''' &= -\frac{L}{2} \left(\frac{\pi}{\beta} \right)^3 \sin \frac{\pi \theta}{\beta}, \quad \ddot{\dot{y}} = -\frac{L}{2} \left(\frac{\pi \omega}{\beta} \right)^3 \sin \frac{\pi \theta}{\beta} \end{aligned} \quad (8.11)$$

The equations for the return from $\theta = 0, y = L$ to $\theta = \beta, y = 0$ are

$$\begin{aligned} y &= \frac{L}{2} \left(1 + \cos \frac{\pi \theta}{\beta} \right) \\ y' &= -\frac{\pi L}{2\beta} \sin \frac{\pi \theta}{\beta}, \quad \dot{y} = -\frac{\pi L \omega}{2\beta} \sin \frac{\pi \theta}{\beta} \end{aligned} \quad (8.12)$$

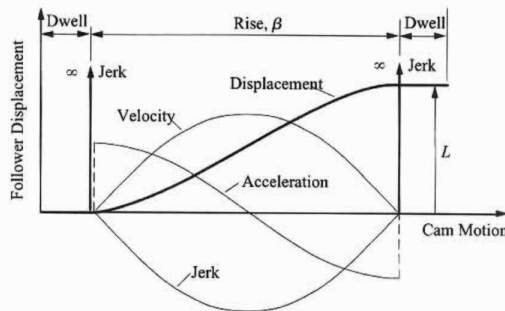


FIGURE 8.16 Shape of the displacement, velocity, acceleration, and jerk curves for simple harmonic motion.

$$y'' = -\frac{L}{2} \left(\frac{\pi}{\beta} \right)^2 \cos \frac{\pi\theta}{\beta}, \quad \ddot{y} = -\frac{L}{2} \left(\frac{\pi\omega}{\beta} \right)^2 \cos \frac{\pi\theta}{\beta}$$

$$y''' = \frac{L}{2} \left(\frac{\pi}{\beta} \right)^3 \sin \frac{\pi\theta}{\beta}, \quad \dddot{y} = \frac{L}{2} \left(\frac{\pi\omega}{\beta} \right)^3 \sin \frac{\pi\theta}{\beta}$$

A simple harmonic displacement diagram can be generated graphically by drawing a semicircle on the vertical axis and dividing it into an even number of segments. The cam motion axis is then divided into the same number of even increments, and horizontal lines are drawn from the points on the semicircle axis. The intersections of the horizontal lines with the corresponding vertical lines give the location of points on the simple harmonic curve. This construction is shown in Fig. 8.17. For the construction, note that

$$\frac{\Delta\alpha}{\Delta\theta} = \frac{180}{\beta}$$

where β is the cam rotation for the follower to move from lift 0 to L . With the advent of computers, the graphical procedure is typically used only for schematic representations of simple harmonic motion.

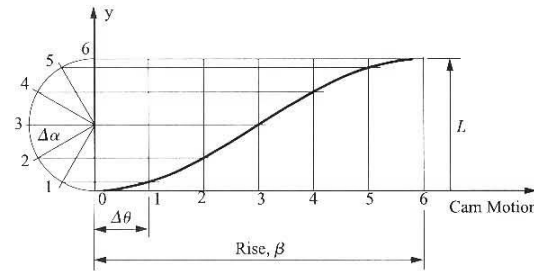


FIGURE 8.17 Graphical construction of displacement diagram for a simple harmonic rise.

EXAMPLE 8.2

Design for Harmonic Motion

Solution

Design a harmonic cam to satisfy the same motion specifications as for Example 8.1. That is, the motion program is to provide a dwell at zero lift for the first 120° of the motion cycle and to dwell at 0.8 in lift for cam angles from 180° to 210° . The cam profile will be laid out using 10° plotting intervals.

The motion specification is as shown in Fig. 8.14 where ϕ is the cam rotation angle.

The rise in the interval 120° to 180° can be computed using Eq. (8.11) if we use $\theta = (\phi - 120^\circ)$ and $0.8 = L$. The resulting expression for the rise is

$$y = \frac{L}{2} \left(1 - \cos \frac{\pi\theta}{\beta} \right) = 0.4 \left(1 - \cos \frac{\pi(\phi - 120^\circ)}{60^\circ} \right) \quad (8.13)$$

The results are given in Table 8.3.

For the return in the interval 210° to 360° , Eq. (8.12) applies if we use $\theta = (\phi - 210^\circ)$ and $0.8 = L$. The resulting expression for the return is

$$y = \frac{L}{2} \left(1 + \cos \frac{\pi\theta}{\beta} \right) = 0.4 \left(1 + \cos \frac{\pi(\phi - 210^\circ)}{150^\circ} \right) \quad (8.14)$$

Using this equation produces the successive values for y given in Table 8.4.

TABLE 8.3 Cam-Follower Data for Rise Using Simple Harmonic Motion in Example 8.2

θ	120°	130°	140°	150°	160°	170°	180°
y	0.0000	0.0536	0.2000	0.4000	0.6000	0.7464	0.8000

TABLE 8.4 Cam-follower Data for Return in Example 8.2

θ	210°	220°	230°	240°	250°	260°	270°	280°
y	0.8000	0.7913	0.7654	0.7236	0.6677	0.6000	0.5236	0.4418
θ	290°	300°	310°	320°	330°	340°	350°	360°
y	0.3582	0.2764	0.2000	0.1323	0.0764	0.0346	0.0087	0.0000

The tabulated lift values may be compared with those of Example 8.1 to observe the differences between comparable parabolic and harmonic transition curves. If plotted, the follower-displacement program would be difficult to distinguish from Fig. 8.15. However, there will be important differences in the values for the derivatives.

8.8 CYCLOIDAL FOLLOWER-DISPLACEMENT PROGRAMS

All of the motions given so far have nonzero values of acceleration (and therefore infinite jerk) at the beginnings and ends of the motion and therefore are limited to relatively low speeds. Cycloidal motion has zero acceleration at the beginning and end of the motion and so is useful for relatively high speeds.

A cycloidal transition produces a sinusoidal acceleration curve. The equations for the rise are

$$y = L \left(\frac{\theta}{\beta} - \frac{1}{2\pi} \sin \frac{2\pi\theta}{\beta} \right)$$

$$y' = \frac{L}{\beta} \left(1 - \cos \frac{2\pi\theta}{\beta} \right), \quad \dot{y} = \frac{L\omega}{\beta} \left(1 - \cos \frac{2\pi\theta}{\beta} \right)$$

$$y'' = \frac{2L\pi}{\beta^2} \sin \frac{2\pi\theta}{\beta}, \quad \ddot{y} = 2L\pi \left(\frac{\omega}{\beta} \right)^2 \sin \frac{2\pi\theta}{\beta}$$

$$y''' = \frac{4L\pi^2}{\beta^3} \cos \frac{2\pi\theta}{\beta}, \quad \ddot{\ddot{y}} = 4L\pi^2 \left(\frac{\omega}{\beta} \right)^3 \cos \frac{2\pi\theta}{\beta}$$

These curves are plotted in Fig. 8.18. There is no discontinuity at the inflection point, and therefore the equations are valid for values of θ from zero to β . The curve is symmetric, and the return is given by $\bar{y} = L - y$. Therefore, $\bar{y}' = -y'$, $\bar{y}'' = -y''$, and $\bar{y}''' = -y'''$.

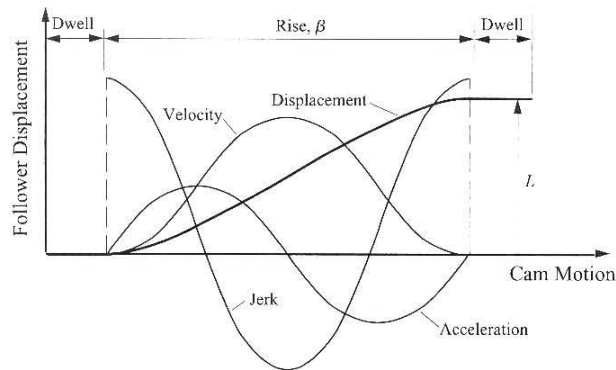


FIGURE 8.18 Shape of the displacement, velocity, acceleration, and jerk relations for cycloidal motion.

Cycloidal motion may be obtained by rolling a circle of radius $L/2\pi$, where L is the total rise, on the displacement axis as shown in Fig. 8.19. However, to construct the curve graphically, a more convenient, alternative way is shown in Fig. 8.19. First, a circle with diameter L/π and center at (β, L) is divided into an even number of increments, and the resulting points are projected onto the displacement axis. The cam motion axis is divided into the same number of increments. A series of lines parallel to the line from the origin to the point (β, L) is then drawn from the projected points on the circle diameter. The intersections of these lines with the corresponding vertical lines from the cam-motion axis give points on the cycloidal curve.

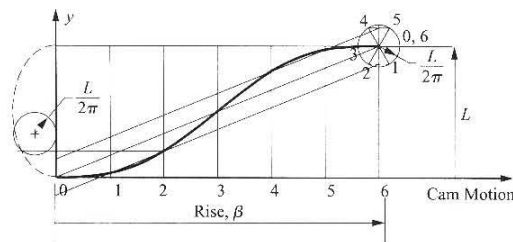


FIGURE 8.19 Graphical construction of displacement diagram for cycloidal motion.

8.9 GENERAL POLYNOMIAL FOLLOWER-DISPLACEMENT PROGRAMS

For high-speed machines, it is common to specify a general polynomial profile for a cam. Depending on the order of the polynomial chosen, it is theoretically possible to match almost any conditions posed by the designer. A polynomial curve is fitted to the rise or return. Only odd-order polynomials are appropriate for rises or returns between dwells if the same conditions are to be matched at both ends of the polynomial. A first-order polynomial gives constant velocity and infinite acceleration at the beginning and end of the transition. This is the uniform motion profile discussed previously. A third-order polynomial gives a parabolic velocity variation, linear acceleration, and infinite jerk at the beginning and end

of the transition. A fifth-order polynomial gives finite acceleration and jerk. The derivative of jerk is infinite at the ends of the transition. A fifth-order fit is the practical maximum unless great care is taken during manufacturing. Dynamic effects resulting from manufacturing errors tend to become more important than those from curve fitting at this stage.

For a general polynomial follower displacement, the displacement function is given by

$$y = f(\theta) = \sum_{i=0}^n A_i \theta^i$$

where θ is the cam angle, and the A 's must be determined from the conditions to be matched. The equation permits us to match the same number of conditions as there are A 's, that is, $n + 1$ conditions. When n is large and the angles are measured in degrees, the terms in the summation can vary greatly in size. For example, if θ is 100° and n is 10, the coefficients of θ^i in the equation can vary hugely, and round-off error will make it difficult to obtain an accurate solution. Therefore, it is convenient to rewrite the displacement equation in terms of the cam rotation angle β , which gives the range over which the equation is to be used. The resulting equation is

$$y = f(\theta) = \sum_{i=0}^n C_i \left(\frac{\theta}{\beta} \right)^i$$

Now the coefficients of the constants are always numbers between 0 and 1, and round-off error problems have been greatly reduced. The constants in the two equations are related by the simple expression

$$A_i = \frac{C_i}{\beta^i}$$

The conditions to be matched will typically involve at least the velocity and acceleration of the follower, and the required equations for these conditions can be written as

$$\dot{y} = \dot{f}(\theta) = \frac{1}{\beta} \frac{d\theta}{dt} \sum_{i=1}^n i C_i \left(\frac{\theta}{\beta} \right)^{(i-1)}$$

and

$$\ddot{y} = \ddot{f}(\theta) = \frac{1}{\beta} \frac{d^2\theta}{dt^2} \sum_{i=1}^n i C_i \left(\frac{\theta}{\beta} \right)^{(i-1)} + \frac{1}{\beta^2} \left(\frac{d\theta}{dt} \right)^2 \sum_{i=2}^n i(i-1) C_i \left(\frac{\theta}{\beta} \right)^{(i-2)}$$

Notice that the summation on the velocity term starts at 1 because C_0 does not appear in the equation, and the summation on the acceleration term starts at 2 because neither C_0 nor C_1 appears in the acceleration equation.

Now if a constant-velocity cam is used,

$$\frac{d\theta}{dt} = \omega$$

and

$$\frac{d^2\theta}{dt^2} = 0$$

where ω is the angular velocity of the cam. The follower equations may then be written as the following:

$$y = f(\theta) = \sum_{i=1}^n C_i \left(\frac{\theta}{\beta} \right)^i$$

$$\dot{y} = \dot{f}(\theta) = \frac{\omega}{\beta} \sum_{i=1}^n i C_i \left(\frac{\theta}{\beta} \right)^{(i-1)}$$

$$\ddot{y} = \ddot{f}(\theta) = \left(\frac{\omega}{\beta} \right)^2 \sum_{i=2}^n i(i-1) C_i \left(\frac{\theta}{\beta} \right)^{(i-2)}$$

$$\ddot{\ddot{y}} = \ddot{\ddot{f}}(\theta) = \left(\frac{\omega}{\beta} \right)^3 \sum_{i=3}^n i(i-1)(i-2) C_i \left(\frac{\theta}{\beta} \right)^{(i-3)}$$

As an example of the use of the polynomial profile, assume that we begin and end the follower displacement with a dwell as shown in Fig. 8.20 and assume that we want to match the position, velocity, and acceleration at both the beginning and end of the period being considered.

For points *A* and *B* in Fig. 8.20, the following conditions apply:

$$\theta = 0 \Rightarrow y = \dot{y} = \ddot{y} = 0$$

$$\theta = \beta \Rightarrow y = L$$

$$\dot{y} = \ddot{y} = 0$$

There are six conditions, so the position equation must have six constants. The resulting equations for position, velocity, and acceleration are

$$y = C_0 + C_1 \left(\frac{\theta}{\beta} \right) + C_2 \left(\frac{\theta}{\beta} \right)^2 + C_3 \left(\frac{\theta}{\beta} \right)^3 + C_4 \left(\frac{\theta}{\beta} \right)^4 + C_5 \left(\frac{\theta}{\beta} \right)^5$$

$$\dot{y} = \frac{\omega}{\beta} \left[C_1 + 2C_2 \left(\frac{\theta}{\beta} \right) + 3C_3 \left(\frac{\theta}{\beta} \right)^2 + 4C_4 \left(\frac{\theta}{\beta} \right)^3 + 5C_5 \left(\frac{\theta}{\beta} \right)^4 \right]$$

$$\ddot{y} = \left(\frac{\omega}{\beta} \right)^2 \left[2C_2 + 6C_3 \left(\frac{\theta}{\beta} \right) + 12C_4 \left(\frac{\theta}{\beta} \right)^2 + 20C_5 \left(\frac{\theta}{\beta} \right)^3 \right]$$

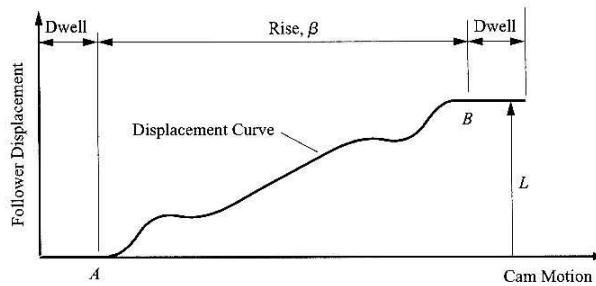


FIGURE 8.20 Initial information for polynomial profile example.

Evaluation of these equations at the beginning and end of the rise period gives the following six equations that must be solved:

$$0 = C_0$$

$$0 = \frac{\omega}{\beta} C_1$$

$$0 = \left(\frac{\omega}{\beta}\right)^2 2C_2$$

$$L = C_0 + C_1 + C_2 + C_3 + C_4 + C_5$$

$$0 = \frac{\omega}{\beta} [C_1 + 2C_2 + 3C_3 + 4C_4 + 5C_5]$$

$$0 = \left(\frac{\omega}{\beta}\right)^2 [2C_2 + 6C_3 + 12C_4 + 20C_5]$$

Solution for the unknown constants C_0 through C_5 gives

$$C_0 = C_1 = C_2 = 0$$

$$C_3 = 10L, \quad C_4 = -15L, \quad C_5 = 6L$$

The displacement equation can then be written in the form

$$y = 10L \left(\frac{\theta}{\beta}\right)^3 - 15L \left(\frac{\theta}{\beta}\right)^4 + 6L \left(\frac{\theta}{\beta}\right)^5$$

This is called the 3-4-5 polynomial transition because of the powers of the terms that remain in the expression. The first three derivatives and the velocity, acceleration, and jerk are given by

$$\begin{aligned} y' &= \frac{30L}{\beta} \left[\left(\frac{\theta}{\beta}\right)^2 - 2\left(\frac{\theta}{\beta}\right)^3 + \left(\frac{\theta}{\beta}\right)^4 \right], & \dot{y} &= \frac{30\omega L}{\beta} \left[\left(\frac{\theta}{\beta}\right)^2 - 2\left(\frac{\theta}{\beta}\right)^3 + \left(\frac{\theta}{\beta}\right)^4 \right] \\ y'' &= \frac{60L}{\beta^2} \left[\left(\frac{\theta}{\beta}\right) - 3\left(\frac{\theta}{\beta}\right)^2 + 2\left(\frac{\theta}{\beta}\right)^3 \right], & \ddot{y} &= 60L \left(\frac{\omega}{\beta}\right)^2 \left[\left(\frac{\theta}{\beta}\right) - 3\left(\frac{\theta}{\beta}\right)^2 + 2\left(\frac{\theta}{\beta}\right)^3 \right] \\ y''' &= \frac{60L}{\beta^3} \left[1 - 6\left(\frac{\theta}{\beta}\right) + 6\left(\frac{\theta}{\beta}\right)^2 \right], & \ddot{\ddot{y}} &= 60L \left(\frac{\omega}{\beta}\right)^3 \left[1 - 6\left(\frac{\theta}{\beta}\right) + 6\left(\frac{\theta}{\beta}\right)^2 \right] \end{aligned}$$

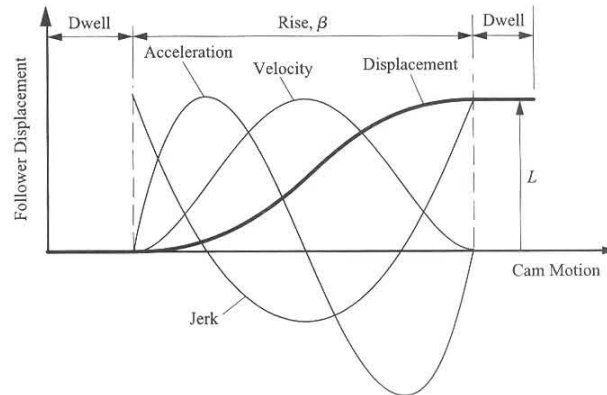


FIGURE 8.21 Shape of the displacement, velocity, acceleration, and jerk relations for the 3-4-5 polynomial motion.

These general relationships are plotted in Fig. 8.21. The displacement results are visually similar to the cycloidal curve, but the velocity, acceleration, and jerk are somewhat different. In general, this type of cam will begin and end its motion more slowly than the other types, and to produce such a cam, extreme machining accuracy is required, especially at the beginning and end of the motion. The machining is commonly done on a computer numerically controlled (CNC) milling machine.

TABLE 8.5 Comparison of the Different Types of Cam-Follower Motion for $\beta = L = 1$

θ	y (linear)	y (parabolic)	y (harmonic)	y (cycloidal)	y (polynomial)
0.00	0.0000	0.0000	0.0000	0.0000	0.0000
0.05	0.0500	0.0050	0.0062	0.0008	0.0012
0.10	0.1000	0.0200	0.0245	0.0065	0.0086
0.15	0.1500	0.0450	0.0545	0.0212	0.0266
0.20	0.2000	0.0800	0.0955	0.0486	0.0579
0.25	0.2500	0.1250	0.1464	0.0908	0.1035
0.30	0.3000	0.1800	0.2061	0.1486	0.1631
0.35	0.3500	0.2450	0.2730	0.2212	0.2352
0.40	0.4000	0.3200	0.3455	0.3065	0.3174
0.45	0.4500	0.4050	0.4218	0.4008	0.4069
0.50	0.5000	0.5000	0.5000	0.5000	0.5000
0.55	0.5500	0.5950	0.5782	0.5992	0.5931
0.60	0.6000	0.6800	0.6545	0.6935	0.6826
0.65	0.6500	0.7550	0.7270	0.7788	0.7648
0.70	0.7000	0.8200	0.7939	0.8514	0.8369
0.75	0.7500	0.8750	0.8536	0.9092	0.8965
0.80	0.8000	0.9200	0.9045	0.9514	0.9421
0.85	0.8500	0.9550	0.9455	0.9788	0.9734
0.90	0.9000	0.9800	0.9755	0.9935	0.9914
0.95	0.9500	0.9950	0.9938	0.9992	0.9988
1.00	1.0000	1.0000	1.0000	1.0000	1.0000

To compare the profiles generated by the different follower-displacement programs, let $\beta = L = 1$, and vary θ from 0 to β . We can then compute y as a function of θ in increments of 0.05. The results are shown in Table 8.5. Notice that the variation among the different profiles is very small in most cases. This emphasizes that extreme accuracy must be achieved if the benefits of using the different follower-displacement programs are to be realized.

8.10 DETERMINING THE CAM PROFILE

Once the follower motion is determined as a function of the cam displacement, the cam surface can be found either graphically or analytically. For extremely accurate cams, the geometry must be determined analytically and the machining must be done using CNC milling machines. For low-speed cams, however, a graphical layout and manual machining are adequate.

In both the graphical and analytical approaches to determining the cam geometry, the cam mechanism must be inverted. That is, the cam is taken as the reference system, and the frame and follower are considered to move relative to the cam. To maintain the correct relative motion, the follower will move relative to the cam in a direction opposite to the motion of the cam relative to the follower.

If we restrict our discussions to planar, rotating cams, four general types of followers are possible: (a) a translating cylindrical-faced follower, (b) a translating flat-faced follower, (c) a rotating cylindrical-faced follower, and (d) a rotating flat-faced follower (Fig. 8.22).

Notice that the cam geometry is independent of the type of joint between the cylindrical-faced follower and the cam. The kinematic design procedure is exactly the same when a roller follower or a solid cylindrical-faced follower is involved. We will consider both graphical and analytical approaches to the design of the cam for each type of follower shown in Fig. 8.22.

8.10.1 Graphical Cam Profile Layout

As already indicated, cam profiles are laid out graphically using inversion. That is, the cam is viewed as stationary, and the successive positions of the follower are located relative to it. This results in a polar plot of successive follower positions. The cam profile is then filled in as the envelope curve of the follower positions.

The first step in laying out the cam profile is to select a base circle radius. The base circle represents the position of the follower at zero lift. Successive lift values are plotted radially outward from the base circle.

Choosing a large base circle radius results in a large cam. However, if the base circle is too small the cam profile may have hollows of smaller radius than the follower. Since the follower will bridge across such a hollow, it will not follow the desired lift program. Obviously, this situation must be avoided, and it is therefore necessary to have a means of computing the radius of curvature of the cam at different locations.

The pressure angle of a cam is the angle between the contact normal and the velocity of the point on the follower at the contact location. Reducing the pressure angle reduces the contact loads and promotes smoother operation with less wear. Increasing the base circle radius decreases the maximum value of pressure angle. Thus, it is good practice to use the largest base circle that the design constraints will allow. As a general rule of thumb, the base circle radius should be two to three times the maximum lift value.

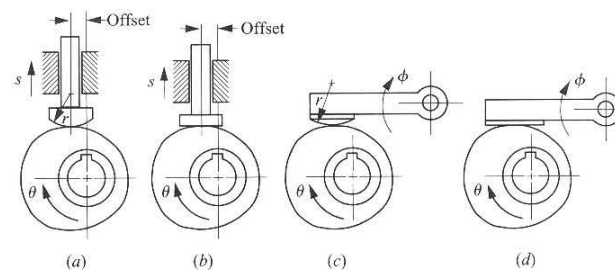


FIGURE 8.22 Common follower configurations for planar, rotating cams. (a) Translating cylindrical-faced follower; (b) translating flat-faced follower; (c) rotating cylindrical-faced follower; (d) rotating flat-faced follower.

EXAMPLE 8.3 Layout of a Cam Profile for a Radial Roller Follower

Layout a cam profile using the harmonic follower displacement profile of Example 8.2. That is, the follower is to dwell at zero lift for the first 120° of the motion cycle and to dwell at 0.8 in lift for cam angles from 180° to 210° . The cam is to have a translating, roller follower with a 1-in roller diameter. The cam will rotate clockwise. Lay out the cam profile using 10° plotting intervals.

Solution

The basic motion specification is as shown in Fig. 8.15. Using the results of Example 8.2 produces the lift values to be plotted given in Table 8.6. Notice that the dwells correspond to locations on the cam where the radius is constant.

TABLE 8.6 Follower Displacements for Example 8.3

θ	0, 360°	10°	20°	30°	40°	50°	60°	70°	80°
y	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
θ	90°	100°	110°	120°	130°	140°	150°	160°	170°
y	0.0000	0.0000	0.0000	0.0000	0.0536	0.2000	0.4000	0.6000	0.7464
θ	180°	190°	200°	210°	220°	230°	240°	250°	260°
y	0.8000	0.8000	0.8000	0.8000	0.7913	0.7654	0.7236	0.6677	0.6000
θ	270°	280°	290°	300°	310°	320°	330°	340°	350°
y	0.5236	0.4418	0.3582	0.2764	0.2000	0.1323	0.0764	0.0346	0.0087

The layout of the cam is accomplished by drawing radial lines at 10° increments. Because the cam rotates clockwise, the radial lines are laid off and labeled in the counterclockwise direction, as shown in Fig. 8.23. Next, the base circle and the prime circle are drawn. The *base circle* is chosen to have a 1.5-in radius, and it is the largest circle that can be drawn inside the cam profile and be tangent to the cam profile. The radius of the *prime circle* is equal to $r_b + r_f$ where r_b is the base circle radius and r_f is the radius of the roller follower. In this problem, the prime-circle radius is 2.0 in. The cam profile is initially laid off from the prime circle to give the pitch curve. The *pitch curve* is the curve traced by the center of the roller follower. Notice that the pitch curve will be the cam profile if r_f is zero. This corresponds to the case of a knife-edged follower.

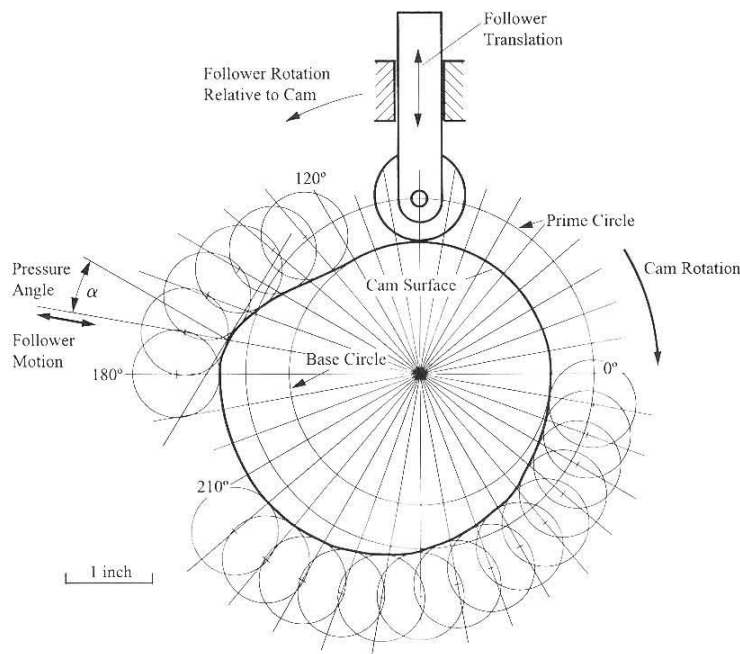


FIGURE 8.23 Layout of the cam profile for Example 8.3. The process of laying out a cam profile is one of inversion. That is, the cam is viewed as being stationary, and successive positions of the follower are plotted relative to it. In this case, a prime circle of 2.0-in radius was chosen. This represents the location of the follower center at zero lift. Positive lift values are plotted outward from the base circle. The successive positions of the follower are then drawn using the plotted points as centers. Finally, the profile is plotted as an envelope curve of the successive follower positions. Because of the inversion, if the cam is to rotate clockwise, the positions of the follower must be plotted in the opposite direction, that is, counterclockwise.

Once the radial lines and prime circle are established, the displacements can be laid off from the prime circle as shown in Fig. 8.23. The radius of the follower is drawn with its center located on the pitch curve at a series of locations. The cam can be defined by drawing a curve tangent to roller locations as shown in Fig. 8.23.

As indicated before, an important parameter for cam motion is the pressure angle. In the case of the translating, roller follower, this is the angle between the direction of the follower travel and the normal to the curve. For a given force on the follower roller, the force in the direction of travel of the follower will be proportional to the cosine of the pressure angle. The force normal to the travel of the follower is proportional to the sine of the pressure angle. Wear on the follower stem will increase with the normal force. Therefore, from design considerations, we want the pressure angle (α) to be as small as possible.

The maximum pressure angle will occur at the *pitch points*. These correspond to the inflection points on the follower displacement curves (see Fig. 8.8). If the torque on the cam is more or less constant, the pressure angles at the pitch points will correspond to the parts of the cycle where the maximum normal force occurs and hence the times when the follower stem wear will be greatest. It will also correspond to the parts of the cycle where the follower will tend to bind in the stem bearing. Because of problems with wear and binding, the pressure angle is usually limited to angles on the order of $\pm 30^\circ$. If the pressure angle becomes excessive, the base circle should be increased or the follower-displacement profile should be changed.

The problem statement indicated that a roller follower was to be designed. However, the construction would be *exactly* the same if a solid cylindrical-faced follower were involved. From the standpoint of the cam geometry, the important issues are the radius of the cylindrical face and the direction of translation relative to the cam.

EXAMPLE 8.4
Layout of a Cam
Profile for a Radial
Flat-Faced
Follower

Solution

Again, lay out a cam profile using the harmonic displacement profile of Examples 8.2 and 8.3. The cam is to have a translating flat-faced follower that is offset by 0.2 in. The cam will rotate clockwise. Lay out the cam profile using 10° plotting intervals.

The basic motion specification is the same as in Example 8.3 (Table 8.6). The layout of the cam is again accomplished by drawing radial lines at 10° increments. Because the cam rotates clockwise, the radial lines are laid off and labeled in the counterclockwise direction as was done in Fig. 8.23. Next the base circle is drawn. Because a flat-faced follower is being designed, there is no prime circle. However, selection of the base circle requires careful consideration.

A major restriction on the cam profile driving a flat-faced follower is that the profile must form a convex surface. This means that the vectors from every point on the cam to the corresponding center of curvature must point toward the interior of the cam. An alternative way to approach the convexity problem is to imagine an arbitrary line drawn across the face of the cam. If it is possible to select an arbitrary line that intersects the cam at more than two points, the cam profile is not convex. If the cam is not convex, the flat-faced follower cannot contact the cam at all points, and the desired motion will not be generated. This condition will be illustrated mathematically when an analytical approach to cam synthesis is discussed. Clearly, the cam generated in Fig. 8.23 does not satisfy the convexity condition; however, this is not necessarily an issue with roller followers. When the resulting cam is not convex for flat-faced followers, we must increase the size of the base circle or change the follower-displacement function. The effect of changing the base circle can be easily investigated by running one of the cam design programs supplied on the disk with this book.

To begin the construction, we can select a base circle somewhat arbitrarily. However, if the radius of curvature at some location on the resulting cam is too small, the base circle diameter must be increased.

The follower is offset, but this does not affect the geometry of the cam. All points on the follower have the same velocity because its motion is pure translation. Therefore, from a kinematic standpoint, the actual location of the follower stem is not important. From a machine design standpoint it is important, however, because the larger the offset, the higher the moment on the follower stem.

We can lay off the displacements in Table 8.6 from the base circle and along the radial lines. At each of these locations on the radial lines, draw a line perpendicular to each radial line. These perpendicular lines correspond to the face of the follower. This is illustrated in Fig. 8.24. The lines for different positions of the follower will form an envelope that defines the geometry of the cam surface. We construct the outline of the cam by drawing a curve that contacts the lines corresponding to the different positions of the follower face at the tangent points.

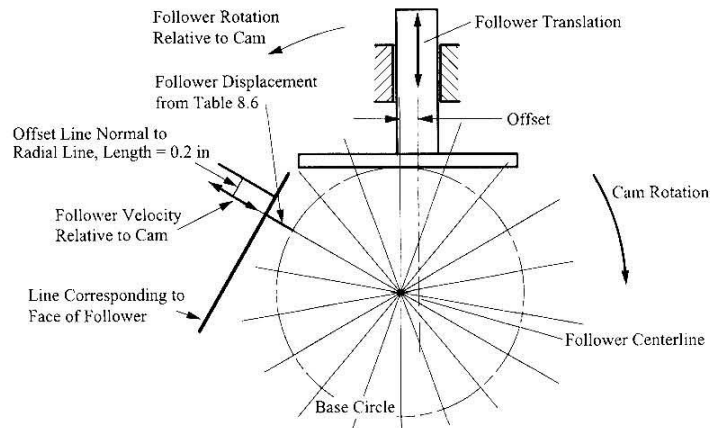


FIGURE 8.24 Basic construction lines for determining the cam profile for a flat-faced follower.

As the lines corresponding to the different positions of the follower face are drawn, successive lines will intersect. For the geometry to be valid, the angle increment for successive intersections must be positive. If an intersection requires a negative angle increment, it will not be possible to generate the cam, and a larger base circle must be used. This situation is illustrated in the current problem in Fig. 8.25 for the positions corresponding to rotation angles of 150° , 160° , 170° , and 180° . In Fig. 8.25, a base circle radius of 1.5 in was chosen. The angle increment for the intersection corresponding to 160° and 170° is positive, but the increment corresponding to 170° and 180° is negative. This situation makes the cam nonconvex and indicates that the base circle is too small and must be increased.

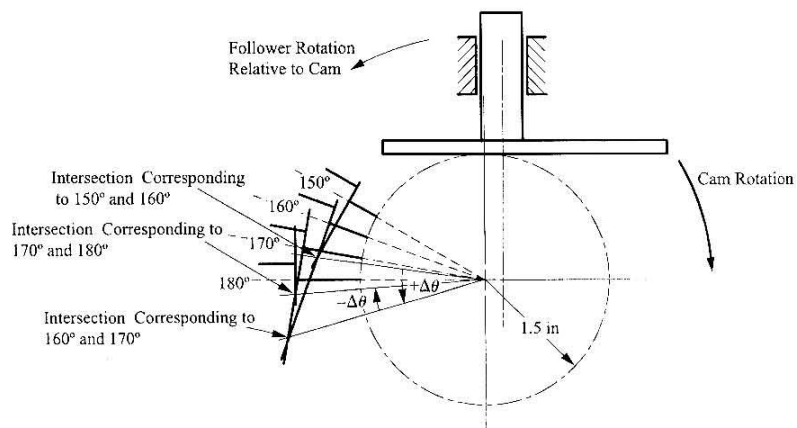


FIGURE 8.25 Condition when the base circle is too small to generate an acceptable cam for a flat-faced follower.

The smallest base circle is the one for which the angle increment corresponding to 170° and 180° is no longer negative, that is, when it is zero. This occurs when the follower-face lines corresponding to 160° , 170° , and 180° intersect. This occurs for a base circle radius of approximately 5.5 in. For this base circle, the cam will have a point or cusp corresponding to the location where the face lines intersect. The envelope of the face lines and the resulting cam is shown in Fig. 8.26. The cam designed for the roller follower and for the same displacement profile is also shown in Fig. 8.26 for comparison. Based on the size of the cam required, a flat-faced follower would not be a good choice for this type of displacement profile.

After the follower-face locations are found, the stem locations can be shown by drawing parallel lines to the radial lines. These are also shown in Fig. 8.26.

As indicated before, an important parameter for cam motion is the pressure angle. When a flat-faced follower is used, the normal to the follower profile is always in the direction of the follower travel if the follower face is perpendicular to the stem. This makes the pressure angle always zero; however, there can be significant lateral loads on the follower bearings caused by the frictional force

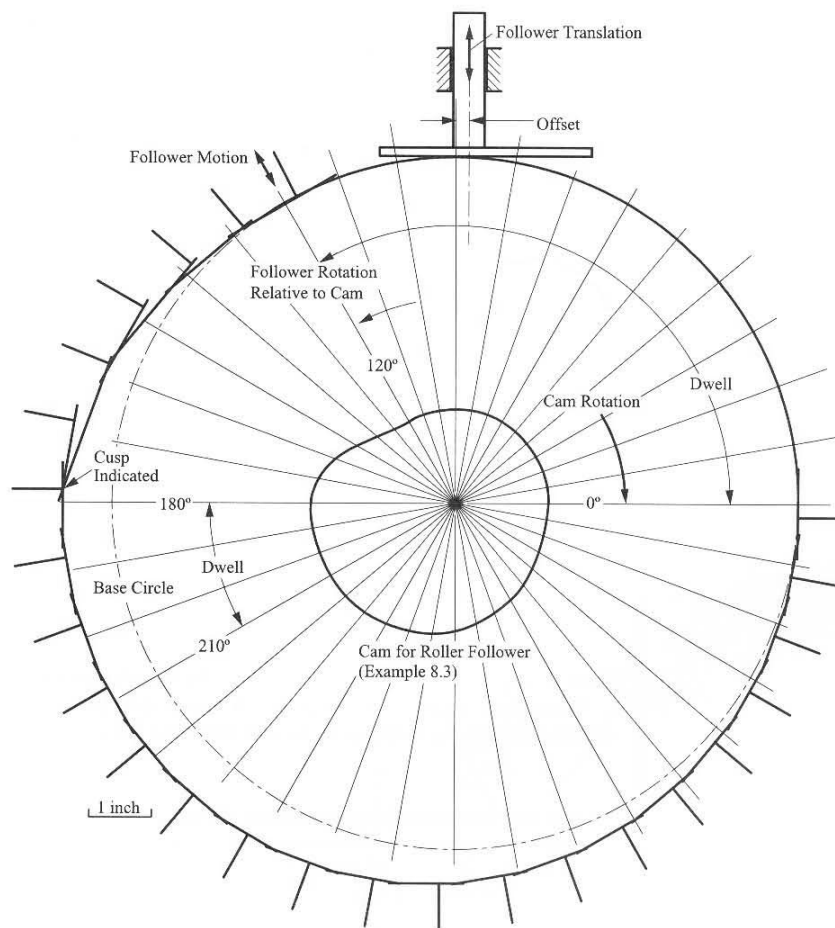


FIGURE 8.26 Envelope generated by drawing lines corresponding to the follower face for Example 8.4. The base circle is the minimum possible for the flat-faced follower to generate the profile indicated in Table 8.6. The cam generated for the roller follower in Example 8.3 is included for comparison.

at the cam–follower interface and by the moment generated by the normal force at the cam–follower interface and the offset line of action. The friction force can be reduced by lubrication but never completely eliminated, and the bearing couple that opposes the moment from the normal force must be addressed in the design of the cam and follower system. Depending on the lubrication and design, the lateral forces can be as high as or higher than the corresponding lateral force with a roller follower. Also, the cam may be so large that, to avoid the convexity condition, the roller follower would be preferred from size considerations.

Another important parameter that must be determined for the design of a cam for a flat-faced follower is the size (length) of the follower face. It is essential that the face be long enough on both sides to maintain contact with the cam on a tangent line. The minimum length of the follower face can be established by direct measurement. The actual length would be equal to the minimum length plus a small increment.

EXAMPLE 8.5
Layout of a Cam
Profile for an
Oscillating
Cylindrical
Follower

Solution

Lay out a cam profile assuming that the follower starts from a dwell for 0° to 120° of cam rotation, and the cam rotates clockwise. The rise occurs during the cam rotation from 120° to 200° . The follower then dwells for 40° of cam rotation, and the return occurs for the cam rotation from 240° to 360° . The amplitude of the follower rotation is 30° , and the follower radius is 0.75 in. Lay out the cam profile using 20° plotting intervals. (This plotting interval is too coarse for the development of an accurate cam; however, it will be used to simplify the resulting drawing.)

The basic motion specification is visually similar to that shown in Fig. 8.15; however, the follower motion is a rotation instead of a translation. To begin the design, we must determine the follower rotation, ϕ , as a function of the cam rotation, θ . Assume parabolic motion for the follower. From section 8.6, the equations for each part of the motion are

$$0 < \theta \leq 120^\circ: \quad \phi = 0$$

$$120^\circ \leq \theta \leq 160^\circ: \quad \phi = 2L \left(\frac{\Delta\theta}{\beta} \right)^2 = 60 \left(\frac{\theta - 120^\circ}{80^\circ} \right)^2$$

$$160^\circ \leq \theta \leq 200^\circ: \quad \phi = L \left[1 - 2 \left(1 - \frac{\Delta\theta}{\beta} \right)^2 \right] = 30 \left[1 - 2 \left(1 - \frac{\theta - 120^\circ}{80^\circ} \right)^2 \right]$$

$$200^\circ \leq \theta \leq 240^\circ: \quad \phi = 30^\circ$$

$$240^\circ \leq \theta \leq 300^\circ: \quad \phi = L \left[1 - 2 \left(\frac{\Delta\theta}{\beta} \right)^2 \right] = 30 \left[1 - 2 \left(\frac{\theta - 240^\circ}{120^\circ} \right)^2 \right]$$

$$300^\circ \leq \theta \leq 360^\circ: \quad \phi = 2L \left(1 - \frac{\Delta\theta}{\beta} \right)^2 = 60 \left(1 - \frac{\theta - 240^\circ}{120^\circ} \right)^2$$

Here, L is treated as the generic amplitude of the follower motion. In this case, L is given as 30° .

Using these equations provides the lift values to be plotted given in Table 8.7. Notice that the dwells correspond to locations on the cam where the radius is constant.

TABLE 8.7 Follower Displacements for Example 8.5

θ	$0, 360^\circ$	20°	40°	60°	80°	100°	120°	140°	160°
y	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	3.7500°	15.0000°
θ	180°	200°	220°	240°	260°	280°	300°	320°	340°
y	26.2500°	30.0000°	30.0000°	30.0000°	28.3333°	23.3333°	15.0000°	6.6666°	1.6666°

The cam-follower system is similar to that shown in Fig. 8.27, and the points shown in Fig. 8.27 will be used to describe the layout of the cam. In particular, A is the location of the axis of rotation of the cam, B is the center of curvature of the follower, and C is the rotation axis of the follower.

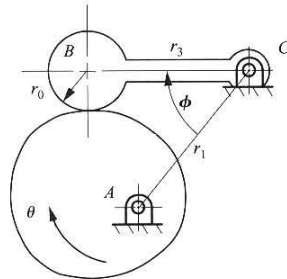


FIGURE 8.27 Cam-follower system to be designed in Example 8.5.

The first step in the cam layout is to draw the base circle and prime circle. The *base circle* is chosen to have a 1.25-in radius, and the radius of the *prime circle* is equal to $r_b + r_0 = 1.25 + 0.75 = 2.0$ in. The base circle radius is the radius of the cam during the dwell for the first 120° of cam rotation.

The second step is to select the distance from the cam rotation axis to the pivot of the follower (AC). The larger the value chosen, the smaller will be the pressure angle; however, this distance also directly affects the size of the cam-follower system. We will choose the distance between pivots to be 4 in. When we invert the motion, the follower pivot will appear to rotate around the cam. Therefore, we must draw a circle with a radius of 4 in about the cam for the follower pivot circle. The pivot circle is shown in Fig. 8.28.

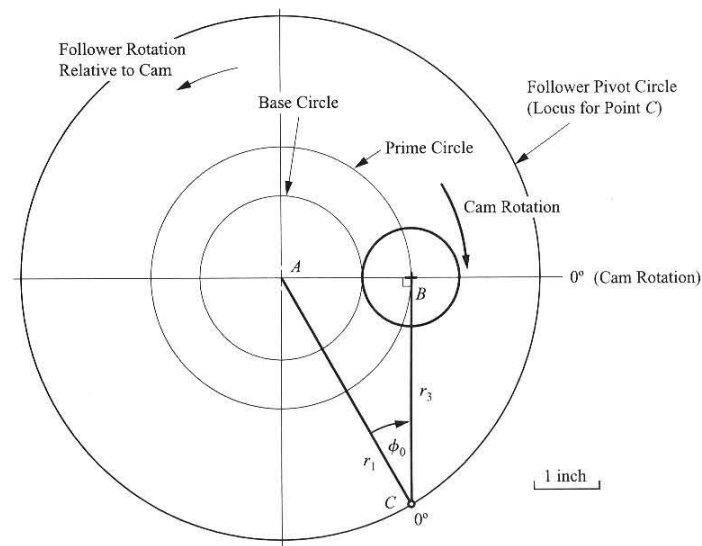


FIGURE 8.28 Location of the follower in the initial position. This determines the reference angle ϕ_0 .

The third step is to determine the length of the follower ($BC = r_3$) and the reference angle ϕ_0 for ϕ . We will locate the follower in the initial dwell position in such a way that the pressure angle is zero. This is done by drawing a line tangent to the prime circle through the ray corresponding to $\theta = 0$. The intersection of this tangent line with the follower pivot circle will give the location of the follower pivot for the initial position of the cam. Two intersections will be given. One will correspond to a CW rotation of the follower, and the second will correspond to a CCW rotation. Based on the problem statement, we will choose the intersection corresponding to the CW rotation as shown in Fig. 8.28. Because a right triangle is involved, the follower length is given by

$$r_3 = \sqrt{r_1^2 - (r_b + r_0)^2} = \sqrt{4^2 - 2^2} = 3.464$$

We also need to determine the base angle, ϕ_0 , because all subsequent displacements of the follower will be measured relative to this angle. If we let the distance between the cam and follower pivots be r_1 , the angle ϕ_0 is given directly by

$$\phi_0 = \tan^{-1} \left(\frac{r_b + r_0}{r_3} \right) = \tan^{-1} \left(\frac{2}{3.464} \right) = 30.000^\circ$$

The location of pivot C for the follower when the cam angle is 0° gives the first position of the follower pivot. Subsequent positions of the pivot will be at angle increments of 20° . Therefore, the fourth step in the cam layout is to draw radial lines at 20° increments from the cam rotation axis to the follower pivot circle starting from the initial position of AC corresponding to the cam rotation angle of 0° . Label the radial lines corresponding to the beginning and end of dwells. These radial lines are shown in Fig. 8.29.

The fifth step is to draw a line tangent to the prime circle from the intersection of the radial lines and the follower pivot circle, as shown in Fig. 8.29. These lines give the position of the follower relative to the cam, if the cam is a simple cylinder. These lines will give the base lines from which to measure the ϕ angles given in Table 8.7. Next lay off lines 3.464-in long at the angles indicated in Table 8.7 from the corresponding base lines. The ends of these lines will be the centers of the cylindrical cam follower in the different positions. Draw circles corresponding to the follower, and construct the cam surface tangent to the follower positions. The final cam is shown in Fig. 8.29.

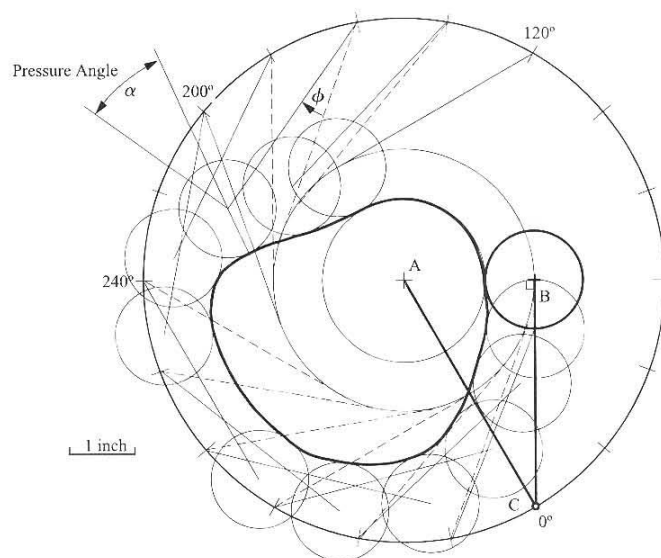


FIGURE 8.29 Final layout of the cam profile for Example 8.5.

As in the case of an axial roller follower, an important parameter for cam motion is the pressure angle. For the oscillating cylindrical follower, this is the angle between the velocity of the contact point on the follower (vector normal to line from contact point and point C) and the normal to the cam at the contact point. To reduce wear on the follower pivot, we want the pressure angle to be as small as possible. In the design shown in Fig. 8.29, the pressure angle will become fairly high ($\alpha > 30^\circ$) in the rise region. To improve the design, the diameter of the base circle should be increased.

The problem statement indicated that a cylindrical follower was to be designed. However, the construction would be *exactly* the same if a roller follower were involved. From the standpoint of the cam geometry, the important issues are the radius of the cylindrical face and the direction of motion relative to the cam.

EXAMPLE 8.6
Layout of the
Cam Profile for
an Oscillating,
Flat-Faced
Follower

Solution

Lay out the rise portion of the cam profile for the follower motion indicated in Table 8.7. The cam will rotate *counterclockwise*. Assume that the follower face angle is 170° , and lay out the cam profile using 20° plotting intervals. Again, this plotting interval is too coarse for an accurate cam; however, it will be used to illustrate the procedure.

The cam-follower system is similar to that shown in Fig. 8.30, and the points shown in that figure will be used to describe the layout of the cam. Point A is the location of the axis of rotation of the cam, C is the rotation axis of the follower, and β is the follower face angle. Point B is the location of the intersection of the centerline of the follower stem with the face of the follower.

The first step in the cam layout is to select and draw the base circle. The base circle is chosen to be 2.0 in.

The second step is to select the distance from the cam rotation axis to the pivot of the follower (AC). The value chosen will affect the size of the cam, and typically the smallest value possible is chosen. The pivot distance must be large enough that the cam does not contact the follower pivot. Also, the force between the cam and follower will increase as the distance decreases. Therefore, it may be necessary to increase the pivot distance from machine design considerations. We will choose the distance between pivots to be 4 in. When we invert the motion, the follower pivot will appear to rotate around the cam. Therefore, we must draw a circle with a radius of 4 in about the cam for the follower pivot circle, as shown in Fig. 8.31.

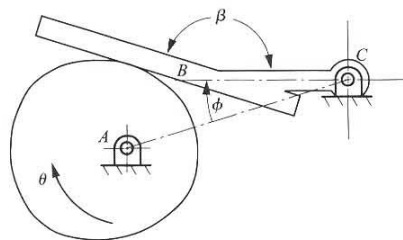


FIGURE 8.30 Cam-follower system to be designed in Example 8.6.

The third step is to determine the reference angle ϕ_0 for ϕ . The location of the follower to determine ϕ_0 will affect the length of the follower and therefore the cost of the system. For simplicity, we will locate the follower in the initial dwell position (cam angle $\theta = 0$) in such a way that point B contacts the cam in that position. This is done by drawing a line tangent to the base circle at B . Then construct a line at an angle of 170° to the tangent line at B . The intersection of this line with the follower pivot circle will locate C in the initial position. This is shown in Fig. 8.31. The length BC can then be computed using the law of cosines. That is,

$$BC^2 + AB^2 = AC^2 + 2(BC)(AB)\cos 100^\circ$$

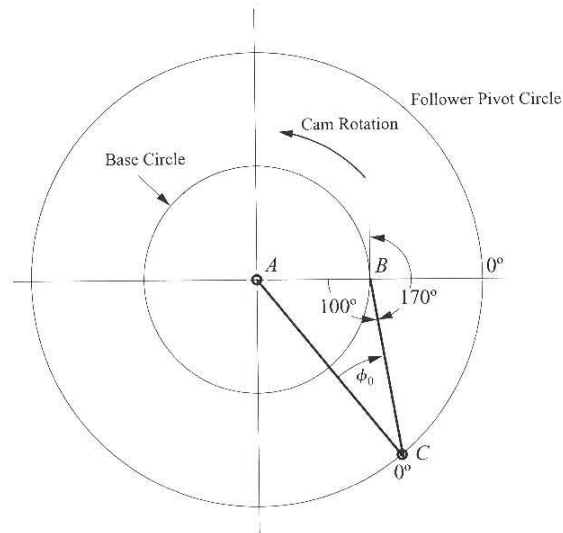


FIGURE 8.31 Determining the initial position of the follower.

or

$$BC^2 - 2(BC)(AB)\cos 100^\circ + (AB^2 - AC^2) = 0$$

and

$$BC = \frac{2(AB)\cos 100^\circ + \sqrt{[2(AB)\cos 100^\circ]^2 - 4(AB^2 - AC^2)}}{2} = \frac{-0.694 + \sqrt{48.482}}{2} = 3.134$$

The base angle, ϕ_0 , can be computed using the law of sines. That is,

$$\phi_0 = \sin^{-1} \left[\frac{AB}{AC} \sin 100^\circ \right] = \sin^{-1} \left[\frac{2}{4} \sin 100^\circ \right] = 29.499^\circ$$

This angle is important because all subsequent displacements of the follower will be measured relative to this angle.

The location of pivot C for the follower when the cam angle is zero gives the first position of the follower pivot. Subsequent positions of the pivot will be at increments of 20° in the clockwise direction (opposite the cam rotation). Therefore, the fourth step in the cam layout is to draw radial lines at 20° increments from the cam rotation axis to the follower pivot circle starting from the initial position of AC corresponding to the cam rotation angle of 0° . These will be the positions of AC for each 20° rotation of the cam. Label the radial lines corresponding to the beginning and end of the first dwell and the second dwell. In the dwell regions, the cam will have a circular contour. These radial lines are shown in Fig. 8.32.

The fifth step is to draw the line CB relative to each position of AC , where the angle between CB and AC is $(\phi_0 + \phi)$ measured in the clockwise direction about C . This will locate the successive positions of B . Next, draw a line through each position of B at an angle of 170° to BC . This will locate the successive positions of the face of the cam follower. Construct the cam surface tangent to the follower positions. The rise portion of the final cam is shown in Fig. 8.32.

Because of the follower face angle β , the pressure angle will not be constant as in the case of the axial flat-faced follower. The pressure angle will depend on the contact point as shown in Fig. 8.32

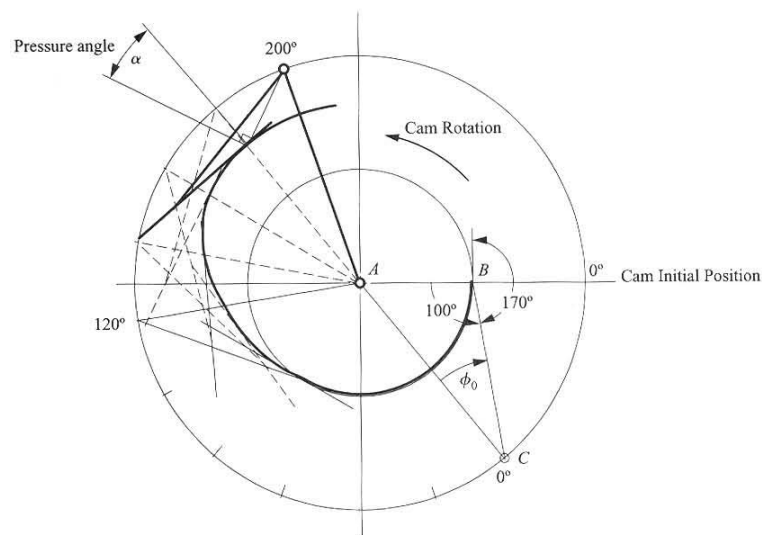


FIGURE 8.32 Final layout of the rise portion of the cam profile for Example 8.6.

and will be equal to $10^\circ \pm \zeta$, where ζ is the angle between a line from the contact point to C and the line BC . If the follower face angle is 180° , the pressure angle will be a constant, that is, 0° . Because of this, the angle β would be 180° unless specific design conditions dictated otherwise.

8.10.2 Analytical Determination of Cam Profile

Although the graphical approach works well for low-speed cams, for high speeds greater accuracy is required, making analytical techniques necessary. The analytical approach to determining the cam profiles also uses inversion. The general procedure is to establish a coordinate system on both the cam and the frame. To determine the location of the follower relative to the cam, it is necessary only to determine the location of the follower relative to the cam coordinate system. Successive positions of the follower will generate an envelope that will define the geometry of the cam profile.

The approach to the design of each of the four different follower systems considered is slightly different, and in each case the analytical approach will follow much the same procedure as was illustrated in the graphical approach.

Analytical Determination of Cam Profile for an Offset, Radial Roller Follower

Prior to determining the geometry of the cam, it is assumed that the basic motion specification for the follower is known in the form $y = f(\theta)$, where y is the displacement of the follower and θ is the cam rotation angle. The base circle radius (r_b), the follower radius (r_f), and the offset δ are also assumed to be known. The positive direction for δ is defined by rotating $R 90^\circ$ in the counterclockwise direction. The procedure is the same for both a roller follower and a cylindrical-faced follower. We will first locate successive positions of the roller center and then determine the envelope formed by the rollers.

Two positions of the follower relative to the cam are shown in Fig. 8.33. In the figure, it is assumed that the cam rotates clockwise, which means that the follower moves relative

to the cam in the counterclockwise direction. The radial position of the center of the follower from the origin of the coordinate system located at the center of the cam is given by

$$R = r_0 + r_b + f(\theta) \quad (8.15)$$

where $f(\theta)$ is the function defining the follower displacement as a function of the cam rotation angle, θ . Referring to Fig. 8.33, we have

$$x = R \cos \theta - \delta \sin \theta = [r_0 + r_b + f(\theta)] \cos \theta - \delta \sin \theta \quad (8.16)$$

and

$$y = R \sin \theta + \delta \cos \theta = [r_0 + r_b + f(\theta)] \sin \theta + \delta \cos \theta \quad (8.17)$$

Given the follower displacement, the center of the roller can be easily computed as a function of the cam rotation angle θ . These equations will also define the cam profile if a knife-edged follower is involved. Also, if the cutter radius is the same as the roller radius, the coordinates can be used to generate the cam profile directly. However, for other cases, we must determine the cam profile indirectly.

As the cam angle θ is incremented, the roller follower can be represented in a series of positions as shown in Fig. 8.34. In any given position, the coordinates of all points on the roller follower can be defined relative to the cam. The cam surface is tangent to the successive positions of the rollers. Therefore, to determine the cam profile, we must locate points on a curve that is tangent to the series of circles.

The series of circles will form an envelope that will define the cam profile. Using envelope theory,^{1,2} it is possible to define the profile exactly; however, envelope theory is somewhat complex except in the case of translating followers. Fortunately, if fine enough increments (less than 1°) are used, the cam profile can be determined very accurately using numerical techniques, and this is the approach that we will use here. The numerical techniques are much simpler than envelope theory, and once the path of the center of the follower is known, the cam profile can be determined without knowledge of whether an oscillating or translating follower was used.

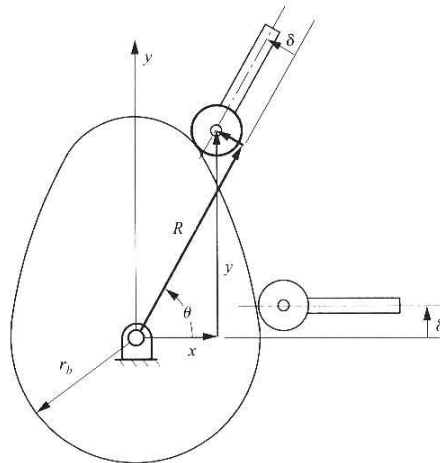


FIGURE 8.33 Two positions of the roller follower relative to the cam.

Referring to Fig. 8.34, we can approximate the cam profile by a series of points defined in a variety of ways. Some of these are as follows:

1. intersections of roller circles (*A*),
2. average tangent location formed by successive tangent lines (*B*),
3. intersections of successive tangent lines (*C*), and
4. tangent points formed by a circle tangent to three successive positions of the roller (*D*).

Other schemes based on fitting other curves to the circles could also be developed. However, it is apparent that each of the procedures will converge to the true tangent points defining the cam profile if the successive positions of the follower are close enough together.

Of the approximate methods available, procedures 2 and 4 will be the most accurate, and the simpler of these is procedure 4. It is also the procedure that is the more accurate of the two. Therefore, this is the procedure that will be developed here. This procedure is especially simple to program, and it can be made as accurate as desired by using increasingly smaller increments of the cam rotation angle. To begin the procedure, we will first determine an approximate location for the center of curvature of the cam as a function of θ .

Cam Radius of Curvature The radius of curvature for a cam is important for several reasons. These include the following:

- If the cam is concave in a given area, the radius of curvature determines the minimum diameter of the cutter that can be used to machine the cam. The radius of curvature of the cam cannot be smaller than the cutter radius if the cam is concave in the area being machined.
- If the cam is concave in a given area, the radius of curvature defines the minimum diameter of the follower that can be used with the cam.

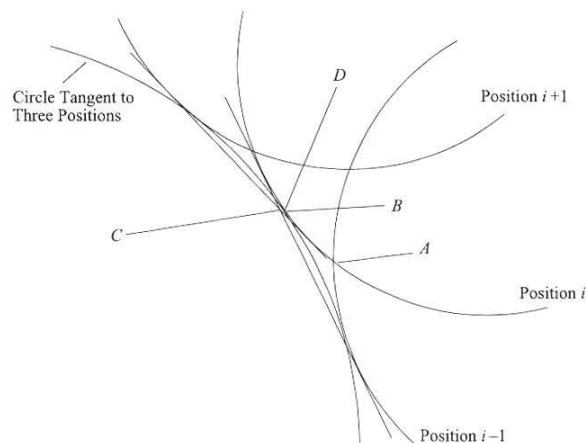


FIGURE 8.34 Approximate procedures for locating a point on the cam profile given three successive positions of the roller follower. Point *A* is found by the simple intersection of two successive positions, point *B* is the average tangent location in position i , point *C* is the intersection of two successive tangents, and point *D* is the tangent location in position i of a circle tangent to positions $i-1$, i , and $i+1$.

- The contact stresses between the cam and the follower are a function of the cam radius of curvature.

If we have a parametric expression for the cam geometry in terms of $x(\theta)$ and $y(\theta)$, an expression for the radius of curvature, ρ , in the x - y plane is given from calculus as

$$\rho = \frac{\sqrt{\left[\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2\right]^3}}{\left(\frac{dx}{d\theta}\right)\left(\frac{d^2y}{d\theta^2}\right) - \left(\frac{dy}{d\theta}\right)\left(\frac{d^2x}{d\theta^2}\right)} \quad (8.18)$$

To determine ρ , we need only differentiate the given expressions for $x(\theta)$ and $y(\theta)$, and substitute the resulting expressions into Eq. (8.18).

The radius of curvature given in Eq. (8.18) will have a magnitude and direction, and both are important. When the cam profile is convex, the radius of curvature is positive, and when the cam is concave, the radius of curvature is negative. The radius can also be zero, which corresponds to a cusp. Technically, the cusp can be concave or convex, but from a practical standpoint, cusps are of interest only for convex surfaces. The transition between concave and convex areas of the cam results in the radius of curvature becoming infinite. The physical interpretations of the various signs of ρ are shown in Fig. 8.35.

Equation (8.18) applies to any cam profile regardless of the type of follower. However, to apply it directly, an analytical expression for $x(\theta)$ and $y(\theta)$ must be available. Except in the case of a translating flat-faced follower, simple expressions for $x(\theta)$ and $y(\theta)$ generally will not be available. Because of this, the purely analytical approach to determining the radius of curvature will not be addressed here except for this special case. Alternatively, a simpler numerical approach will be used.

If the points defining the cam profile are relatively close together, the radius of curvature can be determined numerically. This can be done by evaluating the derivatives in Eq. (8.18) numerically or by fitting circles to the points of the cam profile. Often an approximate value for the radius of curvature is sufficient because in most cases we do not wish to design cams that are close to the operating limits defined by the radii of curvature.

To determine the center of curvature of the cam profile, we will assume that for a given angle θ , the center of curvature of the cam is the same as the center of curvature of the prime curve defined by the locations of the center of the cylindrical follower. We can approximate the center of curvature of the prime by fitting a circle through three successive positions of the center of the cylindrical follower. Designate these positions as $p_{i-1} = (x_{i-1}, y_{i-1})$, $p_i = (x_i, y_i)$, and $p_{i+1} = (x_{i+1}, y_{i+1})$. Also, let the approximate location for the center of curvature be designated as $p_c = (x_c, y_c)$. We can now use Eq. (6.16) in Section 6.3

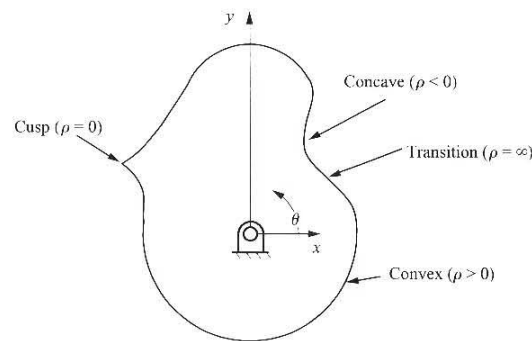


FIGURE 8.35 Interpretation of the sign of ρ .

to find the location of p_c because the procedure for finding p_c is the same as that required to find a pole or centerpoint. The result is

$$\begin{bmatrix} 2(x_{i+1} - x_i) & 2(y_{i+1} - y_i) \\ 2(x_{i-1} - x_i) & 2(y_{i-1} - y_i) \end{bmatrix} \begin{Bmatrix} x_c \\ y_c \end{Bmatrix} = \begin{Bmatrix} (x_{i+1}^2 - x_i^2) + (y_{i+1}^2 - y_i^2) \\ (x_{i-1}^2 - x_i^2) + (y_{i-1}^2 - y_i^2) \end{Bmatrix} \quad (8.19)$$

This linear matrix equation can be solved easily for (x_c, y_c) . The magnitude of the radius of curvature of the prime curve is given by

$$\rho = |r_{p_i/p_c}| = \sqrt{(x_i - x_c)^2 + (y_i - y_c)^2} \quad (8.20)$$

Once the center of curvature is known, the sign of the radius of curvature can be found by taking the cross product of the vector $r_{p_i/p_{i-1}}$ with the vector r_{p_i+1/p_i} . The vectors are shown in Fig. 8.36, and the cross product is given by

$$r_{p_i/p_{i-1}} \times r_{p_{i+1}/p_i} = [(x_i - x_{i-1})(y_{i+1} - y_i) - (x_{i+1} - x_i)(y_i - y_{i-1})]k \quad (8.21)$$

If the cross product is in the positive k direction, the radius of curvature is positive, and the cam is convex at (x_i, y_i) . If the cross product is in the negative k direction, the cam is concave. If the determinant of the coefficient matrix in Eq. (8.19) is zero, then (x_c, y_c) is at infinity, r_{p_i/p_c} is infinite, and an inflection point is indicated. If $(x_c, y_c) = (x_i, y_i)$, the radius of curvature is zero and a cusp is indicated.

Profile Approximation Using Center of Curvature To locate points on the cam profile, we will approximate the center of curvature of three successive positions of the follower as shown in Fig. 8.37. From Eq. (8.19), the center of curvature is approximated as the intersection of lines through the midpoints of the line segments connecting successive follower centers and normal to these line segments as shown in Fig. 8.37.

The corresponding point on the cam profile lies on a line through this center of curvature and the center of the second roller circle. The location of this point (X_i, Y_i) is at the point where this line and the second roller circle intersect. Because there are two such points of intersection, care must be taken to find the right point. For the inner profile the desired point lies on the same side of the follower curve as the cam center.

To locate the line formed by the center of curvature and the second follower center, first calculate the slope orientation angle for the line using

$$\psi_i = \tan^{-1} \left(\frac{y_i - y_c}{x_i - x_c} \right) \quad (8.22)$$

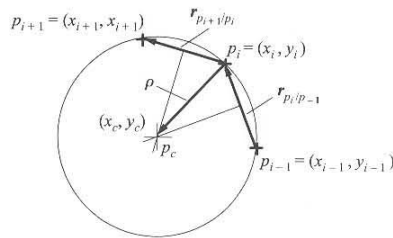


FIGURE 8.36 Use of vectors to determine the sign of ρ .

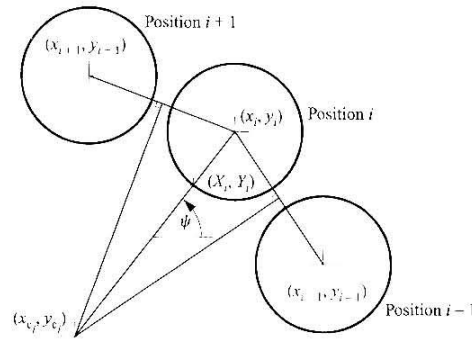


FIGURE 8.37 Approximate center of curvature of roller positions.

This angle is measured from the horizontal axis to the line from the center of curvature of the cam to the contact point. Note that the center of curvature, the contact point, and the center of the cam follower are all collinear. Equation (8.22) can be used directly when the cam is convex. Then the contact point will be located between the center of curvature and the center of the cylindrical follower. However, if the cam surface is concave, the follower center will be located between the center of curvature and the contact point.

The profile can be tested for concavity by solving Eq. (8.21). If the result is positive, the profile is convex and the coordinates on the cam profile are given by

$$\begin{aligned} X_i &= x_i - r_0 \cos \psi_i \\ Y_i &= y_i - r_0 \sin \psi_i \end{aligned} \quad (8.23)$$

If the result is negative, the profile is concave and the coordinates on the cam profile are given by

$$\begin{aligned} X_i &= x_i + r_0 \cos \psi_i \\ Y_i &= y_i + r_0 \sin \psi_i \end{aligned} \quad (8.24)$$

Calculation of the Pressure Angle As indicated previously, the pressure angle is the angle between the outward normal to the cam at the contact point and the direction of the velocity of the follower. The pressure angle is nominally the angle $\phi_i = \psi_i - \theta$, although to use this expression it is necessary to be careful with plus and minus angles. Another way to compute the pressure angle is to locate the center of curvature of the cam relative to the frame coordinate system and draw a line from the center of curvature to the center of the cylindrical follower. Because the cam is assumed to move horizontally, the inclination angle for the line connecting the cam center of curvature with the center of the cylindrical follower will be the pressure angle.

The equations necessary to determine points on the cam profile are summarized in Table 8.8. These can be easily programmed to determine the (X_i, Y_i) coordinates of points on the cam profile, and a MATLAB program for doing this is included on the disk with this book. Given the (X_i, Y_i) coordinates, the cam can be machined on a CNC milling machine. The accuracy of the profile will be determined in part by the angle increment chosen for the cam rotation angle θ .

TABLE 8.8 Summary of Equations for Determining the Cam Profile Coordinates and Pressure Angle for a Translating Cylindrical-Faced or Roller Follower. The Follower Displacement Is Assumed to Be Given by $f(\theta)$, the Radius of the Follower Is r_0 , and the Base Circle Radius Is r_b . There Are Assumed to Be n Points on the Cam Profile. δ Is the Follower Offset.

Radius of curvature

$$\begin{Bmatrix} x_c \\ y_c \end{Bmatrix} = \begin{bmatrix} 2(X_{i+1} - X_i) & 2(Y_{i+1} - Y_i) \\ 2(X_{i-1} - X_i) & 2(Y_{i-1} - Y_i) \end{bmatrix}^{-1} \begin{Bmatrix} (X_{i+1}^2 - X_i^2) + (Y_{i+1}^2 - Y_i^2) \\ (X_{i-1}^2 - X_i^2) + (Y_{i-1}^2 - Y_i^2) \end{Bmatrix}$$

$$\rho = \frac{[(X_i - X_{i-1})(Y_{i+1} - Y_i) - (X_{i+1} - X_i)(Y_i - Y_{i-1})]}{[(X_i - X_{i-1})(Y_{i+1} - Y_i) - (X_{i+1} - X_i)(Y_i - Y_{i-1})]} \sqrt{(X_i - x_c)^2 + (Y_i - y_c)^2}$$

Cam coordinates

$$x_i = [r_0 + r_b + f(\theta_i)] \cos \theta_i - \delta \sin \theta_i$$

$$y_i = [r_0 + r_b + f(\theta_i)] \sin \theta_i + \delta \cos \theta_i$$

$$\psi_i = \tan^{-1} \left(\frac{y_i - y_c}{x_i - x_c} \right) \quad (i = 1, 2, 3, \dots, n)$$

$$X_i = x_i - r_0 \cos \psi_i \quad (\text{convex})$$

$$Y_i = y_i - r_0 \sin \psi_i \quad (\text{convex})$$

$$X_i = x_i + r_0 \cos \psi_i \quad (\text{concave})$$

$$Y_i = y_i + r_0 \sin \psi_i \quad (\text{concave})$$

Pressure angle

$$\phi_i = \psi_i - \theta$$

EXAMPLE 8.7
Cam Profile
Coordinates for
a Radial Roller
Follower

Solution

Determine the cam profile assuming that the follower starts from a dwell from 0° to 90° and rotates CW. The rise occurs with cycloidal motion during the cam rotation from 90° to 180° . The follower then dwells for 60° of cam rotation, and the return occurs with simple harmonic motion for the cam rotation from 240° to 360° . The amplitude of the follower translation is 2 cm, and the follower radius is 1 cm. The base circle radius is 4 cm, and the offset is 0.5 cm.

To solve the problem, we must identify the equations for the follower motion as a function of the cam rotation angle θ , and then select an increment for θ . From Sections 8.7 and 8.8, the equations (expressed in terms of radians) for each part of the motion are

$$\begin{aligned}
0 \leq \theta \leq \pi/2: & \quad y = 0 \\
\pi/2 \leq \theta \leq \pi: & \quad f(\theta) = \frac{2L}{\pi} \left(\left(\theta - \frac{\pi}{2} \right) - \frac{1}{4} \sin 4 \left(\theta - \frac{\pi}{2} \right) \right) \\
\pi \leq \theta \leq 4\pi/3: & \quad f(\theta) = 2, \quad f'(\theta) = f''(\theta) = 0 \\
4\pi/3 \leq \theta \leq 2\pi: & \quad f(\theta) = \frac{L}{2} \left(1 + \cos \frac{3}{2} \left(\theta - \frac{4\pi}{3} \right) \right)
\end{aligned}$$

These equations correspond to $f(\theta)$ in Table 8.8, and using the equations in Table 8.8 we can compute the coordinates of the cam as accurately as we wish. For this problem, an angle increment of 0.1° was used. The problem was solved using the program included on the disk with this book. With this program, it is possible to evaluate quickly the cam profile for areas where the follower roller is too large. The cam, displacement diagram, radius of curvature, and pressure angle plots are shown in Fig. 8.38.

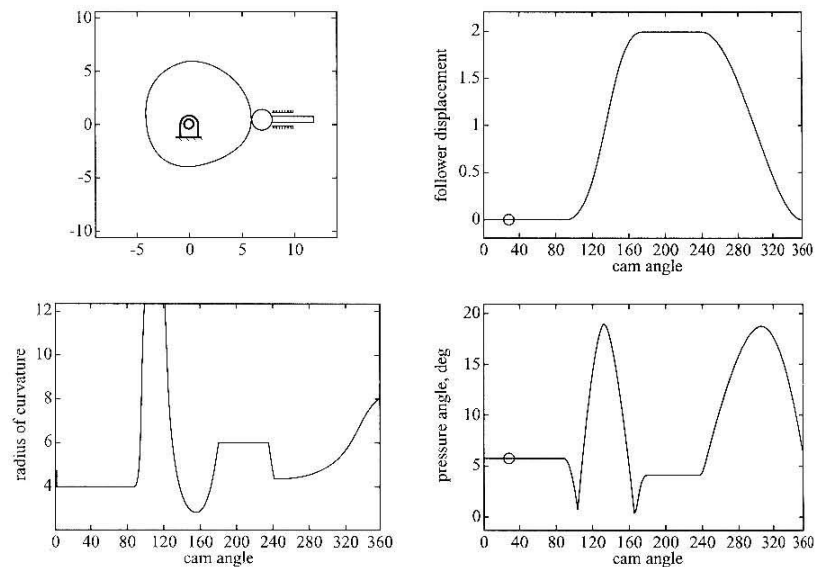


FIGURE 8.38 Cam profile, follower displacement, radius of curvature, and pressure angle for Example 8.7.

Analytical Determination of the Cam Profile for a Radial Flat-Faced Follower The development of the equations for a radial flat-faced follower can be accomplished using a procedure similar to that for a roller follower. Again, we must invert the mechanism so that the follower appears to rotate about the cam.

The radial displacement of the follower relative to the origin of the coordinate system located at the center of the cam is given by

$$R = r_b + f(\theta) \quad (8.25)$$

where $f(\theta)$ is the function defining the follower displacement as a function of the cam rotation angle θ . Referring to Fig. 8.39, we can write Eq. (8.25) as

$$R = y \sin \theta + x \cos \theta$$

If t is the distance from the follower axis to the point of contact,

$$t = y \cos \theta - x \sin \theta = \frac{dR}{d\theta} = f'(\theta)$$

The maximum and minimum values for t will give the minimum limits for the length of the follower face. Note that x and y are defined relative to the coordinate system fixed to the cam. Solving for x and y gives

$$x = R \cos \theta - t \sin \theta$$

$$y = R \sin \theta + t \cos \theta$$

and in terms of $f(\theta)$,

$$\begin{aligned} x &= [r_b + f(\theta)] \cos \theta - f'(\theta) \sin \theta \\ y &= [r_b + f(\theta)] \sin \theta + f'(\theta) \cos \theta \end{aligned} \quad (8.26)$$

In general, we want to use the smallest base circle that will satisfy the geometric constraints. Normally, the base circle is determined in part by the pressure angle; however, for a radial, flat-faced follower, *the pressure angle is always zero*. Therefore, we must select another criterion for determining the base circle radius. The minimum base circle radius will be the one that avoids cusps in the cam profile.

For a given displacement profile, the cam becomes sharper and sharper as the base circle radius decreases. This is shown in Fig. 8.40. Eventually, the cam surface will generate a sharp point or cusp. This condition gives the limiting radius for the base circle.

To establish an equation for the base circle, we can use Eq. (8.18) directly because we have analytical expressions for $x(\theta)$ and $y(\theta)$. For Eq. (8.18), we need the first and second derivatives of $x(\theta)$ and $y(\theta)$. These are given in the following:

$$\frac{\partial x}{\partial \theta} = f'(\theta) \cos \theta - [r_b + f(\theta)] \sin \theta - f''(\theta) \sin \theta - f'(\theta) \cos \theta = -[r_b + f(\theta) + f''(\theta)] \sin \theta$$

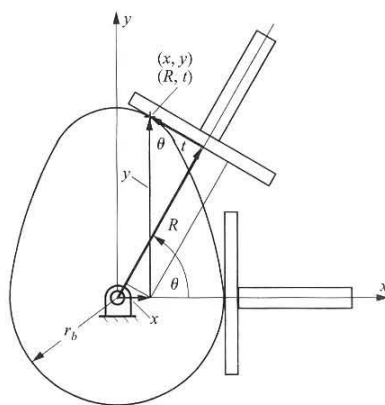


FIGURE 8.39 Radial, flat-faced follower.

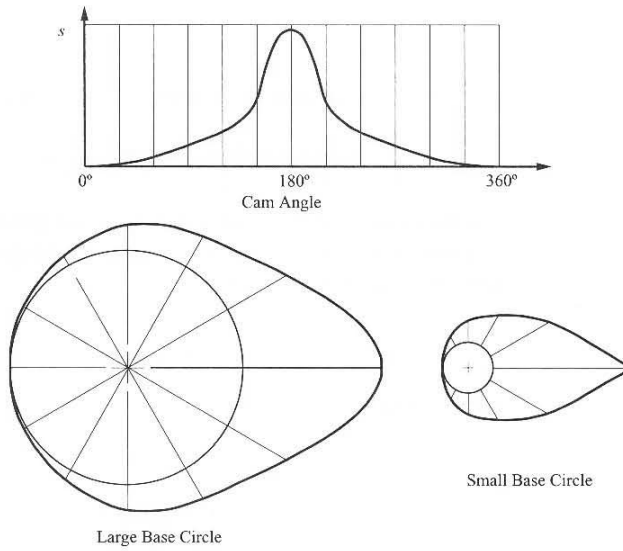


FIGURE 8.40 Formation of a cusp on the surface of the cam.

and

$$\frac{\partial^2 x}{\partial \theta^2} = -[f'(\theta) + f'''(\theta)] \sin \theta - [r_b + f'(\theta) + f'''(\theta)] \cos \theta$$

Also

$$\frac{\partial y}{\partial \theta} = f'(\theta) \sin \theta + [r_b + f(\theta)] \cos \theta + f''(\theta) \cos \theta - f'(\theta) \sin \theta = [r_b + f(\theta) + f''(\theta)] \cos \theta$$

and

$$\frac{\partial^2 y}{\partial \theta^2} = [f'(\theta) + f'''(\theta)] \cos \theta - [r_b + f(\theta) + f''(\theta)] \sin \theta$$

Substitution into Eq. (8.18) gives

$$\rho = \frac{\sqrt{[(dx/d\theta)^2 + (dy/d\theta)^2]^3}}{(dx/d\theta)(d^2y/d\theta^2) - (dy/d\theta)(d^2x/d\theta^2)} = [r_b + f(\theta) + f''(\theta)]$$

For there to be no cusp (and for the cam to be convex) at all locations

$$\rho \geq 0$$

or

$$[r_b + f(\theta) + f''(\theta)] \geq 0 \quad (8.27)$$

for all θ . Both $f(\theta)$ and $f''(\theta)$ will be determined by the follower-displacement schedule. Therefore, only r_b can be externally controlled in Eq. (8.27); that is,

$$r_b \geq -f(\theta) - f''(\theta)$$

If $-f(\theta) - f''(\theta)$ is negative, then any positive or zero value of the base circle radius will be acceptable from a cusp standpoint.

If the cam rotates counterclockwise, the equations developed must be modified slightly. In general, we will treat the follower displacement as a function of the angular displacement rather than the absolute angle. That is, the follower is assumed to translate a given distance regardless of the direction of rotation of the cam. The equations of Sections 8.3–8.9 were derived assuming that the cam angles were positive. When the cam rotates in the counterclockwise direction relative to the frame, the follower will rotate clockwise relative to the cam. This results in negative values for θ . However, the expressions for $f(\theta)$ must use the absolute values of θ . The effect of this is that the correct sign of the derivative $f'(\theta)$ is not preserved for counterclockwise rotation of the cam. The problem is re-solved by changing the sign of the derivative term in Eq. (8.26). That is, when a counterclockwise rotation of the cam is involved, the cam coordinates are given by

$$\begin{aligned} x &= [r_b + f(\theta)] \cos \theta + f'(\theta) \sin \theta \\ y &= [r_b + f(\theta)] \sin \theta - f'(\theta) \cos \theta \end{aligned}$$

The equations necessary to determine points on the cam profile and to determine the length of the follower face are summarized in Table 8.9. These can be programmed easily to determine the (x_i, y_i) coordinates of points on the cam profile relative to the cam coordinate system, and a MATLAB program for doing this is included on the disk with this book. Given the (x_i, y_i) coordinates, the cam can be machined on a CNC milling machine. The accuracy of the profile will be determined in part by the angle increment chosen for the cam rotation angle θ .

TABLE 8.9 Summary of Equations for Determining the Cam Profile Coordinates, Minimum Face Length, and Minimum Base Circle Radius for a Radial Flat-Faced Follower. The Follower Displacement Is Assumed to Be Given by $f(\theta)$, and the Base Circle Radius Is r_b

Cam coordinates—clockwise rotation of cam

$$\begin{aligned} x_i &= [r_b + f(\theta_i)] \cos \theta_i - f'(\theta_i) \sin \theta_i \\ y_i &= [r_b + f(\theta_i)] \sin \theta_i + f'(\theta_i) \cos \theta_i \end{aligned}$$

Cam coordinates—counterclockwise rotation of cam

$$\begin{aligned} x &= [r_b + f(\theta)] \cos \theta + f'(\theta) \sin \theta \\ y &= [r_b + f(\theta)] \sin \theta - f'(\theta) \cos \theta \end{aligned}$$

(continued)

TABLE 8.9 continued

Minimum face length

$$t_{\max, \min} = f'(\theta_i) \Big|_{\max, \min}$$

Radius of curvature

$$\rho = \left[r_b + f(\theta) + f''(\theta) \right]$$

EXAMPLE 8.8
Cam Profile
Coordinates for
Radial Flat-Faced
Follower

Solution

Determine the cam profile for the follower motion given in Example 8.7. First find the minimum base circle radius based on avoiding cusps, and use that base circle to design the cam.

To solve the problem, we must determine the derivatives of the functions for the follower displacement. Because derivatives are involved, the angles in the displacement functions will be converted to radians. For the different intervals, the functions and derivatives are summarized in the following:

$$0 \leq \theta \leq \pi/2: f(\theta) = f'(\theta) = f''(\theta) = 0$$

$$\pi/2 \leq \theta \leq \pi: f(\theta) = \frac{2L}{\pi} \left(\left(\theta - \frac{\pi}{2} \right) - \frac{1}{4} \sin 4 \left(\theta - \frac{\pi}{2} \right) \right), \quad f'(\theta) = \frac{2L}{\pi} \left(1 - \cos 4 \left(\theta - \frac{\pi}{2} \right) \right),$$

$$f''(\theta) = \frac{8L}{\pi} \left(\sin 4 \left(\theta - \frac{\pi}{2} \right) \right)$$

$$\pi \leq \theta \leq 4\pi/3: f(\theta) = 2, \quad f'(\theta) = f''(\theta) = 0$$

$$4\pi/3 \leq \theta \leq 2\pi: f(\theta) = \frac{L}{2} \left(1 + \cos \frac{3}{2} \left(\theta - \frac{4\pi}{3} \right) \right), \quad f'(\theta) = -\frac{3L}{4} \sin \frac{3}{2} \left(\theta - \frac{4\pi}{3} \right),$$

$$f''(\theta) = -\frac{9L}{8} \cos \frac{3}{2} \left(\theta - \frac{4\pi}{3} \right)$$

Given the equations for the follower displacement, we now need only to increment θ and to evaluate the expressions in Table 8.9 to determine the minimum base circle radius and limiting values for the face length. We can then determine the cam coordinates. The values are computed for 10° increments of θ in Table 8.10. From that table, it is clear that the base circle radius must be at least 3.147 cm and the follower face needs to be at least 2.47 cm above the centerline and 1.5 cm below it. Note that a negative value for t implies a distance below the stem centerline, and a positive value implies a distance above the centerline.

The cam will be designed with a base circle radius of 3.2 cm, and the follower face will be 2.6 cm above the stem centerline and 1.6 cm below it. The results obtained from the MATLAB program included on the disk with this book are shown in Fig. 8.41. To illustrate the effect of choosing a base circle radius less than the critical value, the program was rerun with the base circle radius of 1 cm. The results are shown in Fig. 8.42. The result is clearly a cam that cannot be manufactured.

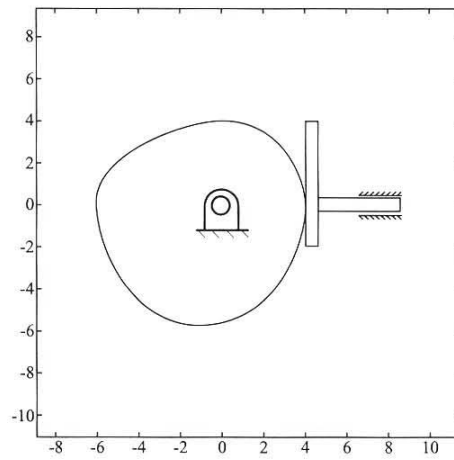


FIGURE 8.41 Cam designed for Example 8.8.

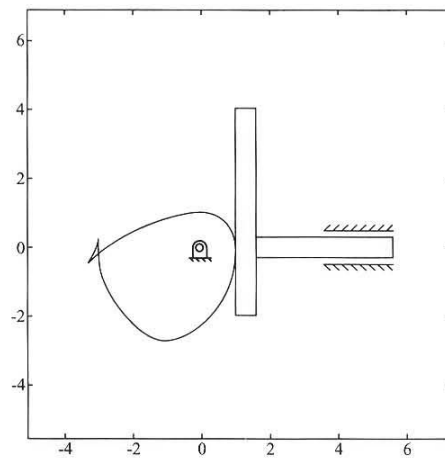


FIGURE 8.42 Results for Example 8.8 when a base circle radius of 1 cm is used. This cam cannot be manufactured. The smallest value for the base circle radius to avoid cusps is 3.147.

TABLE 8.10 Values for $f(\theta)$ and Its Derivatives for Follower Displacement Specified for Example 8.8

θ	$f(\theta)$ (cm)	$f'(\theta)$ (cm)	$f''(\theta)$ (cm)	r_{bmin} (cm)	t (cm)
0°	0.000	0.000	0.000	0.000	0.000
			—dwell—		
90°	0.000	0.000	0.000	0.000	0.000
100°	0.018	0.298	3.274	-3.291	0.298
110°	0.131	1.052	5.016	-5.147	1.052
120°	0.391	1.910	4.411	-4.802	1.910
130°	0.780	2.470	1.742	-2.522	2.470
140°	1.220	2.470	-1.742	0.522	2.470
150°	1.609	1.910	-4.411	2.802	1.910

(continued)

TABLE 8.10 continued

160°	1.869	1.052	-5.016	3.147	1.052
170°	1.982	0.298	-3.274	1.291	0.298
180°	2.000	0.000	0.000	-2.000	0.000
—dwell—					
240°	2.000	-0.000	-2.250	0.250	-0.000
250°	1.966	-0.388	-2.173	0.207	-0.388
260°	1.866	-0.750	-1.949	0.083	-0.750
270°	1.707	-1.061	-1.591	-0.116	-1.061
280°	1.500	-1.299	-1.125	-0.375	-1.299
290°	1.259	-1.449	-0.582	-0.676	-1.449
300°	1.000	-1.500	0.000	-1.000	-1.500
310°	0.741	-1.449	0.582	-1.324	-1.449
320°	0.500	-1.299	1.125	-1.625	-1.299
330°	0.293	-1.061	1.591	-1.884	-1.061
340°	0.134	-0.750	1.949	-2.083	-0.750
350°	0.034	-0.388	2.173	-2.207	-0.388

Analytical Determination of a Cam Profile for an Oscillating, Cylindrical Follower

The equations for an oscillating, cylindrical follower can be derived on the basis of the graphical procedure. Again, we must invert the mechanism so that the follower appears to rotate about the cam. To begin the procedure, we will assume that the cam base-circle radius r_b , the radius r_0 of the cylindrical follower, the pivot distance r_1 , and the distance r_3 from the follower pivot to the center of the cylindrical contour are known.

In the procedure, the prime curve will be located first. This is the path traced by the center of curvature of the cylindrical follower. The cam profile can then be determined using the procedure developed in the first subsection of Section 8.10.2. The follower is assumed to begin in its lowest position when the follower contacts the base circle on the cam, as shown in Fig. 8.43. For this position, we need to determine the initial angles, θ_0 and ϕ_0 , where θ_0 gives the initial angle for the cam and ϕ_0 gives the initial angle for the follower. The motion of the cam and follower will be measured relative to these initial angles, respectively. From Fig. 8.43a, these angles can be computed using the law of cosines. That is,

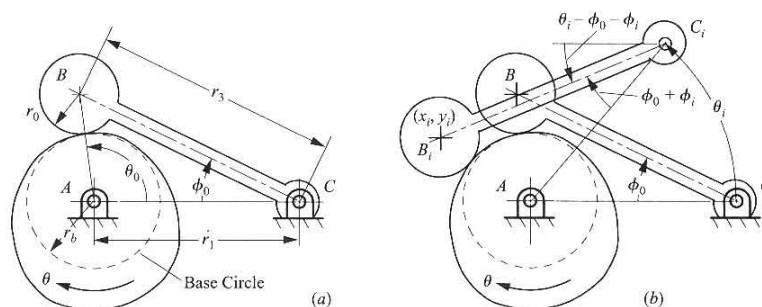


FIGURE 8.43 Oscillating, roller-faced follower. (a) Initial geometry. (b) Displaced geometry.

$$\phi_0 = \cos^{-1} \left[\frac{r_1^2 + r_3^2 - (r_b + r_0)^2}{2r_1r_3} \right] \quad (8.28)$$

and

$$\theta_0 = \cos^{-1} \left[\frac{r_1^2 + (r_b + r_0)^2 - r_3^2}{2r_1(r_b + r_0)} \right] \quad (8.29)$$

To compute the coordinates of the center of the follower (B) relative to a coordinate system at point A on the cam, we will assume that the cam rotates clockwise so that the follower rotates *counterclockwise* relative to the cam.

Referring to Fig. 8.43b, for a given cam rotation angle θ , we have that the coordinates of point B_i are given by

$$x_i = r_1 \cos \theta_i + r_3 \cos[\pi + \theta_i - \phi_i - \phi_0] \quad (8.30)$$

and

$$y_i = r_1 \sin \theta_i + r_3 \sin[\pi + \theta_i - \phi_i - \phi_0] \quad (8.31)$$

Given the coordinates of B_i for a series of cam rotation angles (θ_i), we can compute the coordinates of the corresponding contact points (X_i, Y_i) on the cam face using the procedure in Section 8.10.2.

To compute the pressure angle at any given position, we must find the angle between a vector in the direction of the follower travel and a vector that is normal to the follower roller at each location, as shown in Fig. 8.44. The direction of travel of the follower pivot (point C) relative to the cam is simply θ the cam rotation angle (see Fig. 8.44); therefore, a vector in the direction of travel of point B relative to the cam is

$$\mathbf{a} = 1 \angle \gamma_i$$

where

$$\gamma_i = \theta_i - \phi_i - \phi_0 + \pi/2$$

A vector in the direction normal to the roller is the line from (X_i, Y_i) to (x_i, y_i); therefore, the resulting vector is

$$\mathbf{b} = 1 \angle \lambda_i$$

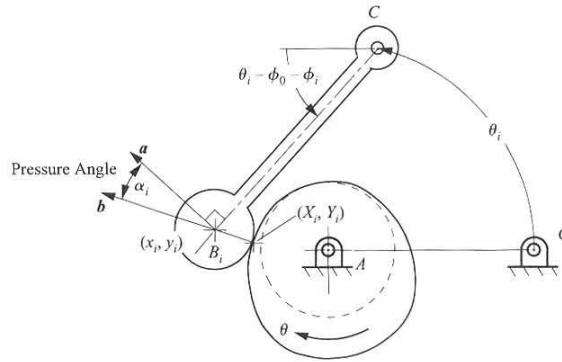


FIGURE 8.44 Pressure angle for oscillating, roller-faced follower.

where

$$\lambda_i = \tan^{-1} \left[\frac{y_i - Y_i}{x_i - X_i} \right]$$

The resulting pressure angle is simply the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$. That is,

$$\alpha_i = \cos^{-1} \left[\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \right] = \cos^{-1} [\cos \gamma_i \cos \lambda_i + \sin \gamma_i \sin \lambda_i] \quad (8.32)$$

The equations to determine points on the cam profile are summarized in Table 8.11. In Table 8.11, the equations for computing (X_i, Y_i) and for the radius of curvature of the cam are taken from Table 8.8. These can be programmed easily to determine the (X_i, Y_i) coordinates of points on the cam profile, and a MATLAB program for doing this is included on the disk with this book. Given the (X_i, Y_i) coordinates, the cam can be machined on a CNC milling machine. The accuracy of the profile will be determined in part by the increment chosen for the cam rotation angle θ .

TABLE 8.11 Summary of Equations for Determining the Cam Profile Coordinates and Pressure Angle for an Oscillating Cylindrical-Faced or Roller Follower. The Follower Oscillation ϕ Is Assumed to Be Given by $f(\theta)$, the Radius of the Follower Is r_0 , the Base Circle Radius Is r_b , the Distance Between the Cam and Follower Pivots Is r_1 , and the Length of the Follower Is r_3 . There Are Assumed to Be n Points on the Cam Profile

Radius of curvature

$$\begin{aligned} \begin{Bmatrix} x_c \\ y_c \end{Bmatrix} &= \begin{bmatrix} 2(X_{i+1} - X_i) & 2(Y_{i+1} - Y_i) \\ 2(X_{i-1} - X_i) & 2(Y_{i-1} - Y_i) \end{bmatrix}^{-1} \begin{Bmatrix} (X_{i+1}^2 - X_i^2) + (Y_{i+1}^2 - Y_i^2) \\ (X_{i-1}^2 - X_i^2) + (Y_{i-1}^2 - Y_i^2) \end{Bmatrix} \\ \rho &= \frac{[(X_i - X_{i-1})(Y_{i+1} - Y_i) - (X_{i+1} - X_i)(Y_i - Y_{i-1})]}{[(X_i - X_{i-1})(Y_{i+1} - Y_i) - (X_{i+1} - X_i)(Y_i - Y_{i-1})]} \sqrt{(X_i - x_c)^2 + (Y_i - y_c)^2} \end{aligned}$$

Cam coordinates

$$\begin{aligned} \phi_0 &= \cos^{-1} \left[\frac{r_1^2 + r_3^2 - (r_b + r_0)^2}{2r_1r_3} \right] \\ x_i &= r_1 \cos \theta_i + r_3 \cos[\pi + \theta_i - \phi_i - \phi_0]; \quad y_i = r_1 \sin \theta_i + r_3 \sin[\pi + \theta_i - \phi_i - \phi_0] \\ \psi_i &= \tan^{-1} \left(\frac{y_i - y_c}{x_i - x_c} \right) \quad (i = 1, 2, 3, \dots, n) \end{aligned}$$

(continued)

TABLE 8.11 continued

$$\begin{aligned} X_i &= x_i - r_0 \cos \psi_i \quad (\text{convex}) \\ Y_i &= y_i - r_0 \sin \psi_i \end{aligned}$$

$$\begin{aligned} X_i &= x_i + r_0 \cos \psi_i \quad (\text{concave}) \\ Y_i &= y_i + r_0 \sin \psi_i \end{aligned}$$

Pressure angle

$$\gamma_i = \theta_i - \phi_i - \phi_0 + \pi/2$$

$$\lambda_i = \tan^{-1} \left(\frac{y_i - Y_i}{x_i - X_i} \right)$$

$$\alpha_i = \cos^{-1} [\cos \gamma_i \cos \lambda_i + \sin \gamma_i \sin \lambda_i]$$

EXAMPLE 8.9
Cam Profile
Coordinates for a
Roller Follower
That Oscillates

Solution

Determine the cam profile assuming that the follower dwells while the cam rotates *counterclockwise* from 0° to 90° . The rise occurs with 3-4-5 polynomial motion during the cam rotation from 90° to 180° . The follower then dwells for 90° of cam rotation, and the return occurs with simple harmonic motion for the cam rotation from 270° to 360° . The amplitude of the follower oscillation is 30° , and the follower radius is 1 in. The base circle radius is 2 in and the distance between pivots is 6 in. The length of the follower is to be determined such that the pressure angle starts out at zero.

To solve the problem, we must identify the equations for the follower motion as a function of the cam rotation angle θ and then select an increment for θ . Because the cam is rotating counterclockwise, the follower rotates clockwise relative to the cam. Mathematically, this is accomplished by using negative angles for the displacements starting from 0. However, the equations developed for the follower displacements give the follower position as a function of the cam position and do not depend on the direction of cam rotation. For a given angular displacement, the rise will be the same whether the cam is to be designed for clockwise or counterclockwise rotation. Therefore, we must use the magnitude of the cam angle from the start of the interval over which the function is defined. In the equations, we will let $\bar{\theta}$ be the magnitude of θ . Then from Sections 8.9 and 8.7, the equations (expressed in terms of radians) for each part of the motion become

$$0 < \bar{\theta} \leq \pi/2: \phi = 0$$

$$\begin{aligned} \pi/2 \leq \bar{\theta} \leq \pi \quad (\beta = \pi/2; L = \pi/6): \phi &= 10L \left(\frac{\bar{\theta} - \pi/2}{\beta} \right)^3 - 15L \left(\frac{\bar{\theta} - \pi/2}{\beta} \right)^4 + 6L \left(\frac{\bar{\theta} - \pi/2}{\beta} \right)^5 \\ &= \left(\frac{5\pi}{3} \right) \left(\frac{\bar{\theta} - \pi/2}{\pi/2} \right)^3 - \left(\frac{5\pi}{2} \right) \left(\frac{\bar{\theta} - \pi/2}{\pi/2} \right)^4 + \pi \left(\frac{\bar{\theta} - \pi/2}{\pi/2} \right)^5 \end{aligned}$$

$$\pi \leq \bar{\theta} \leq 3\pi/2: \phi = \frac{\pi}{6}$$

$$3\pi/2 \leq \bar{\theta} \leq 2\pi \quad (\beta = \pi/2; L = \pi/6): \phi = \frac{L}{2} \left(1 + \cos \frac{\pi}{\beta} \left(\bar{\theta} - \frac{3\pi}{2} \right) \right) = \frac{\pi}{12} \left(1 + \cos 2 \left(\bar{\theta} - \frac{3\pi}{2} \right) \right)$$

To determine the length, r_3 , of the follower that will give zero pressure angle in the initial position, refer to Fig. 8.44. From that figure, the angle between the $(r_0 + r_b)$ and r_3 must be 90° for the pressure angle to be zero during a dwell. Therefore,

$$r_3 = \sqrt{r_1^2 - (r_b + r_0)^2} = \sqrt{6^2 - (2 + 1)^2} = 5.196 \text{ in}$$

The equations for ϕ and r_3 can be used to determine ϕ_0 in Table 8.11, and using the equations in Table 8.11 we can compute the coordinates of the cam as accurately as we wish by using a small increment for the cam rotation angle θ .

For the calculations, the angle θ is incremented in the negative direction (the follower rotates CW relative to the cam). Therefore, the points corresponding to the center of the roller (prime curve points) will be generated and labeled initially in the CW direction. However, the equations in Table 8.11 were derived assuming that the points are ordered in the CCW direction. Therefore, after all of the prime points are generated, they must be reordered in the CCW direction. After the prime points are reordered in this way, the corresponding points on the cam surface can be computed directly as was done in Example 8.7.

For plotting purposes, we have used the program on the disk included with this book with an angle increment of 1° , which is adequate for visual purposes. If we want to machine the cam with high accuracy, a finer increment for θ would be used. It should be noted, however, that when very fine increments are used, the errors of the computations using MATLAB can be significant. The cam, displacement diagram, radius of curvature, and pressure angle plots are shown in Fig. 8.45.

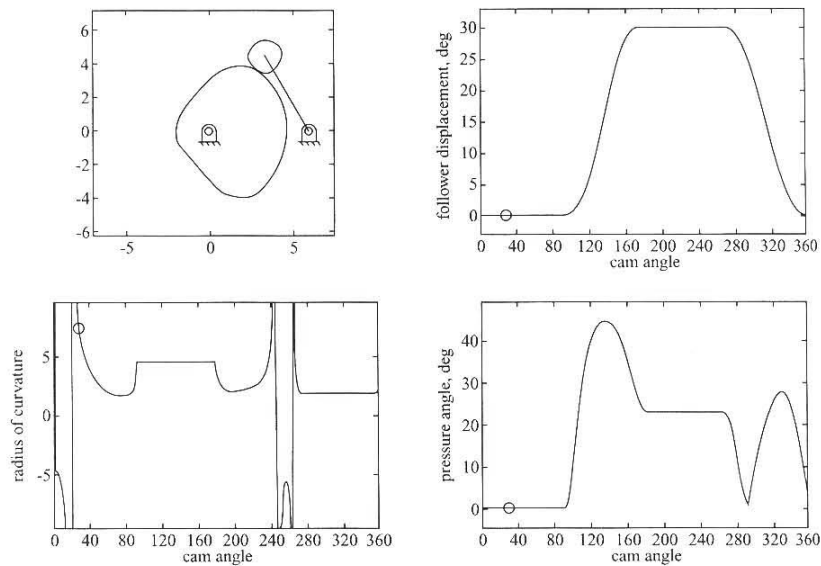


FIGURE 8.45 Cam profile, follower displacement, radius of curvature, and pressure angle for Example 8.9.

Analytical Determination of a Cam Profile for a Flat-Faced Follower That Oscillates

The cam-follower system is shown in Fig. 8.46, and the points shown in that figure will be used to describe the design of the cam. Point A is the location of the axis of rotation of the cam, C is the axis of rotation of the follower, and d is the follower offset. Point B is the location of the tangent point between the face of the follower and the cam surface, and D is located at the intersection of the follower face and a line from C perpendicular to the follower face. The distance R is the perpendicular distance from the follower face to the center of rotation of the cam.

The development of the equations for an oscillating flat-faced follower is similar to the graphical procedure in Example 8.6. As in the other cases, we must invert the mechanism so that the follower appears to rotate about the cam. To begin the procedure, we will assume that the cam base radius r_b , the pivot distance r_1 , and the follower angle β are known.

In the procedure, we will locate the line representing the face of the cam first. We will then locate the points on the cam by locating successive circles that are tangent to three lines.

The first step is to determine the reference angle ϕ_0 that gives the orientation of the follower in the initial position. As shown in Fig. 8.46, in this position,

$$R = r_b = AB$$

From the geometry in Fig. 8.47, we have

$$AE = \frac{r_1}{1 + d/R} = \frac{r_1}{1 + d/r_b}$$

Then

$$BE = \sqrt{AE^2 - r_b^2}$$

and

$$\phi_0 = \tan^{-1} \frac{r_b}{BE}$$

For each position of the cam given by θ , we will know $\phi(\theta)$ from the follower-displacement equations. The orientation of the follower face relative to the cam is then given by the angle $\phi + \phi_0$.

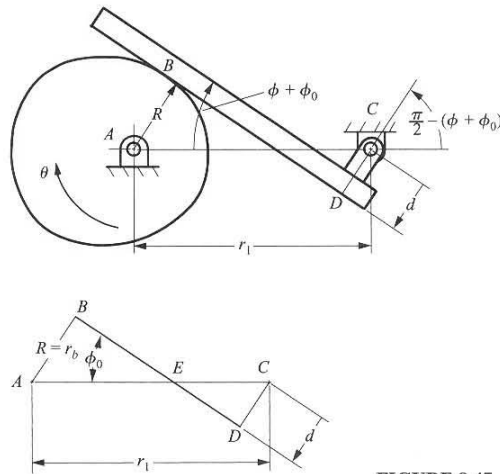


FIGURE 8.46 Geometry for oscillating, flat-faced follower.

FIGURE 8.47 Geometry for finding ϕ_0 .

To locate the follower relative to the cam as the cam rotates, we can invert the motion relative to the cam. For simplicity, we will again assume that the cam rotates CW relative to the frame, which means that the frame rotates CCW relative to the cam. A typical position is shown in Fig. 8.48.

For a given rotation of the cam, the coordinates of point C relative to the cam are

$$x_C = r_1 \cos \theta$$

and

$$y_C = r_1 \sin \theta$$

Similarly, the coordinates of point D are

$$x_D = x_C + d \cos \left(\frac{3\pi}{2} - \phi - \phi_0 + \theta \right)$$

and

$$y_D = y_C + d \sin \left(\frac{3\pi}{2} - \phi - \phi_0 + \theta \right)$$

To locate the line defining the follower face, we need to locate a line oriented at an angle of $\theta_i - \phi_i - \phi_0$ and passing through D_i . The equation of the line defining the follower face can be defined parametrically in terms of the distance τ from D_i as shown in Fig. 8.49. The parametric equations are

$$x_f = x_D + \tau \cos \zeta \quad (8.33)$$

and

$$y_f = y_D + \tau \sin \zeta \quad (8.34)$$

where

$$\zeta = \theta - \phi - \phi_0$$

Using Eqs. (8.33) and (8.34), we can define a line corresponding to the follower face for each position chosen for the cam. To define the cam profile, we must locate a series of points that are tangent to the follower face in each position. The coordinates of the tangent points can be approximated using the geometry shown in Fig. 8.50. Assume that we have

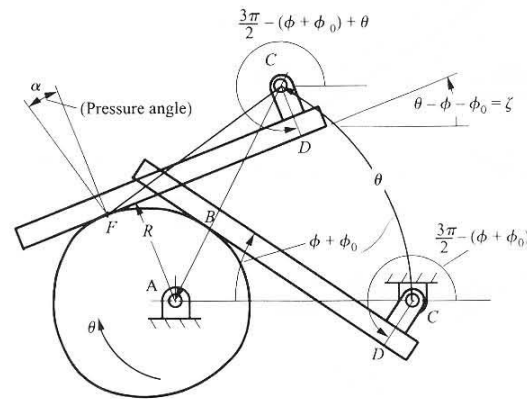


FIGURE 8.48 Motion of an oscillating flat-faced follower relative to a cam.

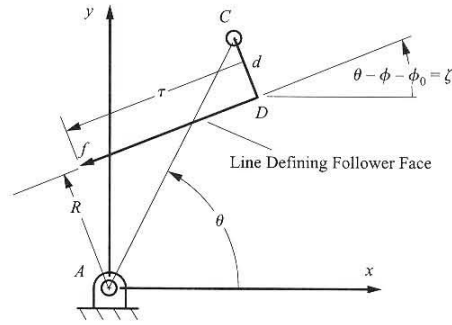


FIGURE 8.49 Parametric line defining a follower face. Here, τ would be negative.

three positions ($i-1$, i , and $i+1$) of the follower relative to the cam. Each follower face line can be defined in terms of the coordinates of D and τ and ζ . To determine the location of the cam contact point on the centerline, approximate the cam by a circle that is tangent to each of the lines simultaneously. Referring to Fig. 8.50, we see that this can be done as follows:

1. Find the successive intersections (G_i and G_{i+1}) of the face lines. To find G_i , use Eqs. (8.33) and (8.34), set $x_{f_i} = x_{f_{i-1}}$ and $y_{f_i} = y_{f_{i-1}}$, and solve the resulting set of linear equations for τ_{i-1} and τ_i :

$$\begin{Bmatrix} x_{D_i} - x_{D_{i-1}} \\ y_{D_i} - y_{D_{i-1}} \end{Bmatrix} = \begin{bmatrix} -\cos \zeta_i & \cos \zeta_{i-1} \\ -\sin \zeta_i & \sin \zeta_{i-1} \end{bmatrix} \begin{Bmatrix} \tau_i \\ \tau_{i-1} \end{Bmatrix}$$

The location of G_i is then given by

$$x_{G_i} = x_{D_i} + \tau_i \cos \zeta_i$$

and

$$y_{G_i} = y_{D_i} + \tau_i \sin \zeta_i$$

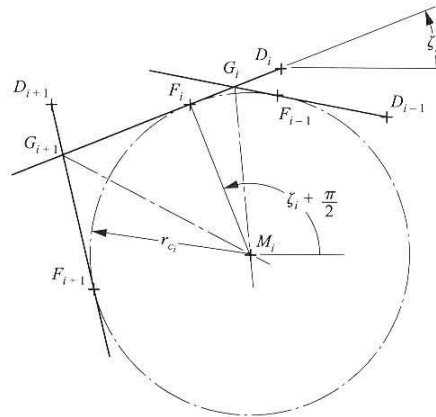


FIGURE 8.50 Locating a circle that is tangent to three successive positions of the follower face relative to the cam.

2. Find the equation of the line that bisects the angle between positions i and $i - 1$ of the face line. This line passes through point G_i and is oriented at an angle of

$$\gamma_i = \frac{\zeta_i + \zeta_{i-1}}{2} + \frac{3\pi}{2}$$

The parametric equation of the line is then given by

$$x_{m_i} = x_{G_i} + v_i \cos \gamma_i$$

and

$$y_{m_i} = y_{G_i} + v_i \sin \gamma_i$$

where v_i is the distance from G_i in the positive γ_i direction.

3. Find the intersection of the lines bisecting the angles at G_i and G_{i+1} . For this, set $x_{m_i} = x_{m_{i+1}}$ and $y_{m_i} = y_{m_{i+1}}$ and solve the resulting set of linear equations for v_i and v_{i+1} .

$$\begin{Bmatrix} x_{G_i} - x_{G_{i+1}} \\ y_{G_i} - y_{G_{i+1}} \end{Bmatrix} = \begin{bmatrix} -\cos \gamma_i & \cos \gamma_{i+1} \\ -\sin \gamma_i & \sin \gamma_{i+1} \end{bmatrix} \begin{Bmatrix} v_i \\ v_{i+1} \end{Bmatrix}$$

The coordinates of the center of the circle tangent to the three positions of the face line are then given by

$$x_{M_i} = x_{G_i} + v_i \cos \gamma_i$$

and

$$y_{M_i} = y_{G_i} + v_i \sin \gamma_i$$

4. Find the radius of the tangent circle and coordinates of the tangent points. These are obtained by locating a line through M_i and perpendicular to each face line. The coordinates of the point F_i are given by

$$x_{F_i} = x_{M_i} + r_{c_i} \cos \left(\zeta_i + \frac{\pi}{2} \right) = x_{G_i} + \eta_i \cos \zeta_i$$

and

$$y_{F_i} = y_{M_i} + r_{c_i} \sin \left(\zeta_i + \frac{\pi}{2} \right) = y_{G_i} + \eta_i \sin \zeta_i$$

where η_i is the distance from G_i to the tangency point. Both r_{c_i} and η_i can be found by solving the resulting set of simultaneous linear equations:

$$\begin{Bmatrix} x_{M_i} - x_{G_i} \\ y_{M_i} - y_{G_i} \end{Bmatrix} = \begin{bmatrix} -\cos \left(\zeta_i + \frac{\pi}{2} \right) & \cos \zeta_i \\ -\sin \left(\zeta_i + \frac{\pi}{2} \right) & \sin \zeta_i \end{bmatrix} \begin{Bmatrix} r_{c_i} \\ \eta_i \end{Bmatrix} \quad (8.35)$$

The location of the tangent point is then given by

$$x_{F_i} = x_{G_i} + \eta_i \cos \zeta_i \quad (8.36)$$

and

$$y_{F_i} = y_{G_i} + \eta_i \sin \zeta_i \quad (8.37)$$

To find the tangency points on the other faces, substitute $i - 1$ for i and then $i + 1$ for i in Eqs. (8.36) and (8.37).

5. The cam coordinates can be approximated by the points F_i computed in step 4. To determine the minimum face length, we can compute the distance from F_i to D_i for each position of the cam.

To compute the pressure angle at any given position, we must find the angle between a vector in the direction of the follower travel and a vector that is normal to the follower face at each location as shown in Fig. 8.48. The direction of travel of the follower is given by a unit vector $\mathbf{a}$ perpendicular to line $F_i C_i$. This vector is given by

$$\mathbf{a} = 1 \angle \psi_i$$

where

$$\psi_i = \tan^{-1} \left[\frac{y_{F_i} - y_{C_i}}{x_{F_i} - x_{C_i}} \right] - \frac{\pi}{2}$$

We can approximate the direction that is perpendicular to the follower face by a unit vector $\mathbf{b}$ from M_i to F_i or

$$\mathbf{b} = 1 \angle \lambda_i$$

where

$$\lambda_i = \tan^{-1} \left[\frac{y_{F_i} - y_{M_i}}{x_{F_i} - x_{M_i}} \right] = \zeta_i + \frac{\pi}{2}$$

The resulting pressure angle is simply the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$. That is,

$$\alpha_i = \cos^{-1} \left[\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \right] = \cos^{-1} [\cos \psi_i \cos \lambda_i + \sin \psi_i \sin \lambda_i] \quad (8.38)$$

The equations necessary to determine points on the cam profile are summarized in Table 8.12, and the equations for the pressure angle and radius of curvature of the cam are summarized in Table 8.13. The equations for the radius of curvature are taken from Table 8.8. The equations in Table 8.12 can be easily programmed to determine the coordinates of points on the cam profile, and a MATLAB program for doing so is included on the disk with this book. Given the coordinates of a set of sequential points on the profile, the cam can be machined on a CNC milling machine. The accuracy of the profile will be determined in part by the increment chosen for the cam rotation angle θ .

TABLE 8.12 Summary of Equations for Determining the Cam Profile Coordinates for an Oscillating Flat-Faced Follower. The Follower Oscillation ϕ Is Assumed to Be Given By $f(\theta)$, the Follower Offset Is d , the Base Circle Radius Is r_b , and the Distance between the Cam and Follower Pivots Is r_1 . There Are Assumed to Be n Points on the Cam Profile

Cam coordinates

$$AE = \frac{r_1}{1 + d/r_b}, \quad BE = \sqrt{AE^2 - r_b^2}$$

$$\phi_0 = \tan^{-1} \frac{r_b}{BE}$$

$$x_{C_i} = r_1 \cos \theta_i, \quad y_{C_i} = r_1 \sin \theta_i$$

$$x_{D_i} = x_{C_i} + d \cos \left(\frac{3\pi}{2} - \phi_i - \phi_0 + \theta_i \right), \quad y_{D_i} = y_{C_i} + d \sin \left(\frac{3\pi}{2} - \phi_i - \phi_0 + \theta_i \right)$$

$$\zeta_i = \theta_i - \phi_i - \phi_0$$

$$\begin{Bmatrix} x_{D_i} - x_{D_{i-1}} \\ y_{D_i} - y_{D_{i-1}} \end{Bmatrix} = \begin{bmatrix} -\cos \zeta_i & \cos \zeta_{i-1} \\ -\sin \zeta_i & \sin \zeta_{i-1} \end{bmatrix} \begin{Bmatrix} \tau_i \\ \tau_{i-1} \end{Bmatrix} \quad (i = 2, 3, \dots, n)$$

$$\begin{Bmatrix} x_{D_1} - x_{D_n} \\ y_{D_1} - y_{D_n} \end{Bmatrix} = \begin{bmatrix} -\cos \zeta_1 & \cos \zeta_n \\ -\sin \zeta_1 & \sin \zeta_n \end{bmatrix} \begin{Bmatrix} \tau_1 \\ \tau_n \end{Bmatrix}$$

$$x_{G_i} = x_{D_i} + \tau_i \cos \zeta_i, \quad y_{G_i} = y_{D_i} + \tau_i \sin \zeta_i$$

$$\gamma_i = \frac{\zeta_i + \zeta_{i-1}}{2} + \frac{3\pi}{2}$$

$$\begin{Bmatrix} x_{G_i} - x_{G_{i+1}} \\ y_{G_i} - y_{G_{i+1}} \end{Bmatrix} = \begin{bmatrix} -\cos \gamma_i & \cos \gamma_{i+1} \\ -\sin \gamma_i & \sin \gamma_{i+1} \end{bmatrix} \begin{Bmatrix} v_i \\ v_{i+1} \end{Bmatrix} \quad (i = 1, 2, \dots, n-1)$$

$$\begin{Bmatrix} x_{G_n} - x_{G_1} \\ y_{G_n} - y_{G_1} \end{Bmatrix} = \begin{bmatrix} -\cos \gamma_n & \cos \gamma_1 \\ -\sin \gamma_n & \sin \gamma_1 \end{bmatrix} \begin{Bmatrix} v_n \\ v_1 \end{Bmatrix}$$

$$x_{M_i} = x_{G_i} + v_i \cos \gamma_i, \quad y_{M_i} = y_{G_i} + v_i \sin \gamma_i$$

$$\begin{Bmatrix} x_{M_i} - x_{G_i} \\ y_{M_i} - y_{G_i} \end{Bmatrix} = \begin{bmatrix} -\cos(\zeta_i + \pi/2) & \cos \zeta_i \\ -\sin(\zeta_i + \pi/2) & \sin \zeta_i \end{bmatrix} \begin{Bmatrix} r_{c_i} \\ \eta_i \end{Bmatrix}$$

$$x_{F_i} = x_{G_i} + \eta_i \cos \zeta_i, \quad y_{F_i} = y_{G_i} + \eta_i \sin \zeta_i$$

TABLE 8.13 Summary of Equations for Determining the Pressure Angle and Radius of Curvature for an Oscillating Flat-Faced Follower**Pressure angle**

$$\psi_i = \tan^{-1} \left[\frac{y_{F_i} - y_{C_i}}{x_{F_i} - x_{C_i}} \right] - \frac{\pi}{2}$$

$$\lambda_i = \tan^{-1} \left[\frac{y_{F_i} - y_{M_i}}{x_{F_i} - x_{M_i}} \right]$$

$$\alpha_i = \cos^{-1} [\cos \psi_i \cos \lambda_i + \sin \psi_i \sin \lambda_i]$$

Radius of curvature

$$\begin{aligned} \begin{Bmatrix} x_c \\ y_c \end{Bmatrix} &= \begin{bmatrix} 2(X_{i+1} - X_i) & 2(Y_{i+1} - Y_i) \\ 2(X_{i-1} - X_i) & 2(Y_{i-1} - Y_i) \end{bmatrix}^{-1} \begin{Bmatrix} (X_{i+1}^2 - X_i^2) + (Y_{i+1}^2 - Y_i^2) \\ (X_{i-1}^2 - X_i^2) + (Y_{i-1}^2 - Y_i^2) \end{Bmatrix} \\ \rho &= \frac{\left[(X_i - X_{i-1})(Y_{i+1} - Y_i) - (X_{i+1} - X_i)(Y_i - Y_{i-1}) \right]}{\left[(X_i - X_{i-1})(Y_{i+1} - Y_i) - (X_{i+1} - X_i)(Y_i - Y_{i-1}) \right]} \sqrt{(X_i - x_c)^2 + (Y_i - y_c)^2} \end{aligned}$$

EXAMPLE 8.10**Cam Profile
Coordinates for a
Flat-Faced
Follower That
Oscillates****Solution**

Assume that the follower starts from a dwell from 0° to 45° and rotates CW. The rise occurs with simple harmonic motion during the cam rotation from 45° to 180° . The follower then dwells for 90° of cam rotation, and the return occurs with simple harmonic motion for the cam rotation from 270° to 360° . The amplitude of the follower oscillation is 20° , and the follower offset is 0.5 in. The base circle radius is 2 in, and the distance between pivots is 6 in.

To solve the problem, we must specify the length of the follower face, identify the equations for the follower motion as a function of the cam rotation angle θ , and then select an increment for θ . The length of the follower face is somewhat arbitrary as long as it is large enough to maintain contact with the cam. The minimum value can be calculated by computing the distance F_i to D_i for each position of the follower relative to the cam. However, in this example, we will select the length to be 9 in, which is large enough to ensure that the follower will maintain contact with the cam.

The second part of the displacement schedule is similar to that used in Example 8.9, and the equation for the rise portion can be obtained from Section 8.7. The resulting equations (expressed in terms of radians) are as follows:

$$0 \leq \theta \leq \pi/2: \phi = 0$$

$$\pi/4 \leq \theta \leq \pi \left(\beta = 3\pi/4; L = \pi/9 \right): \phi = \frac{L}{2} \left(1 - \cos \frac{\pi}{\beta} \left(\theta - \frac{\pi}{4} \right) \right) = \frac{\pi}{18} \left(1 - \cos \frac{4}{3} \left(\theta - \frac{\pi}{4} \right) \right)$$

$$\pi \leq \theta \leq 3\pi/2: \phi = \frac{\pi}{6}$$

$$3\pi/2 \leq \theta \leq 2\pi \left(\beta = \pi/2; L = \pi/9 \right): \phi = \frac{L}{2} \left(1 + \cos \frac{\pi}{\beta} \left(\theta - \frac{3\pi}{2} \right) \right) = \frac{\pi}{18} \left(1 + \cos 2 \left(\theta - \frac{3\pi}{2} \right) \right)$$

Given the equations for the follower displacement, we now need only increment θ and evaluate the expressions in Table 8.12 and 8.13 to determine the cam coordinates and pressure angle. The values were computed for 2° increments of θ , and the results obtained from the MATLAB program included on the disk with this book are shown in Fig. 8.51.

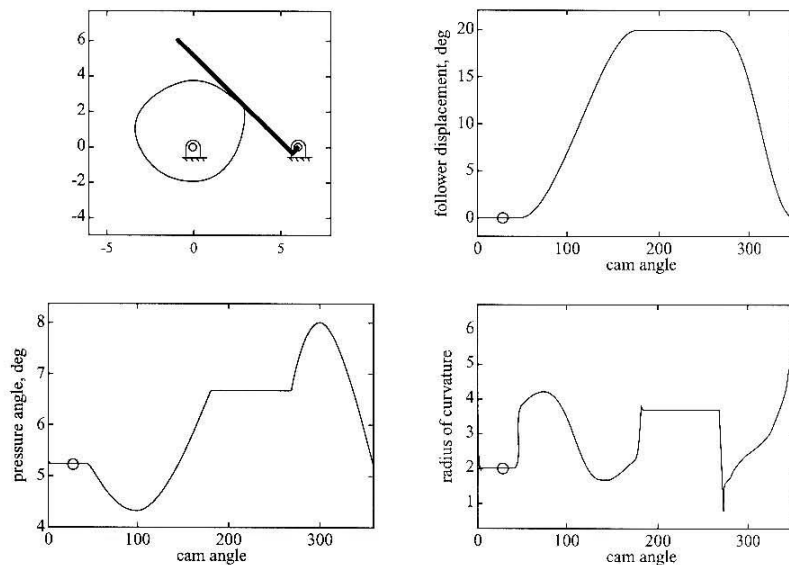


FIGURE 8.51 Cam profile, follower displacement, radius of curvature, and pressure angle for Example 8.10.

It is much more difficult to design a cam for an oscillating flat-faced follower than for the other types of followers. For example, if the amplitude of oscillation in Example 8.10 is changed to 30° , a cam will be developed that has a discontinuity and therefore cannot be manufactured. Changing the base circle radius may not always improve the situation, and the extent to which the base circle can be changed is limited by the distance between pivots. The problem can be improved by reducing the amplitude of oscillation, by changing the functions chosen for the rise and return, or by increasing the ranges chosen for the rise and return. However, in some cases, it may be necessary to choose another type of follower.

REFERENCES

¹Chen, F. Y., *Mechanics and Design of Cam Mechanisms*, Pergamon Press, New York (1982).

²Wilson, C. E., and Sadler, J. P. *Kinematics and Dynamics of Machinery*, Harper Collins, New York (1993).

PROBLEMS

EXERCISE PROBLEMS ON FOLLOWER-DISPLACEMENT SCHEDULES

8.1 A cam that is designed for cycloidal motion drives a flat-faced follower. During the rise, the follower displaces 1 in for 180° of cam rotation. If the cam angular velocity is constant at 100 rpm, determine the displacement, velocity, and acceleration of the follower at a cam angle of 60° .

8.2 A constant-velocity cam is designed for simple harmonic motion. If the flat-faced follower displaces 2 in for 180° of cam rotation and the cam angular velocity is 100 rpm, determine the displacement, velocity, and acceleration when the cam angle is 45° .

8.3 A cam drives a radial, knife-edged follower through a 1.5-in rise in 180° of cycloidal motion. Give the displacement at 60° and 100° . If this cam is rotating at 200 rpm, what are the velocity (ds/dt) and the acceleration (d^2s/dt^2) at $\theta = 60^\circ$?

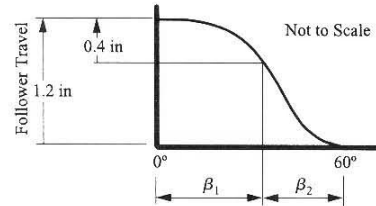
8.4 Draw the displacement schedule for a follower that rises through a total displacement of 1.5 in with constant acceleration for $1/4$ revolution, constant velocity for $1/8$ revolution, and constant deceleration for $1/4$ revolution of the cam. The cam then dwells for $1/8$ revolution, and returns with simple harmonic motion in $1/4$ revolution of the cam.

8.5 Draw the displacement schedule for a follower that rises through a total displacement of 20 mm with constant acceleration for $1/8$ revolution, constant velocity for $1/4$ revolution, and constant deceleration for $1/8$ revolution of the cam. The cam then dwells for $1/4$ revolution and returns with simple harmonic motion in $1/4$ revolution of the cam.

8.6 Draw the displacement schedule for a follower that rises through a total displacement of 30 mm with constant acceleration for 90° of rotation and constant deceleration for 45° of cam rotation. The follower returns 15 mm with simple harmonic motion in 90° of cam rotation and dwells for 45° of cam rotation. It then returns the remaining 15 mm with simple harmonic motion during the remaining 90° of cam rotation.

8.7 Draw the displacement schedule for a follower that rises through a total displacement of 3 in with cycloidal motion in 120° of cam rotation. The follower then dwells for 90° and returns to 0° with simple harmonic motion in 90° of cam rotation. The follower then dwells for 60° before repeating the cycle.

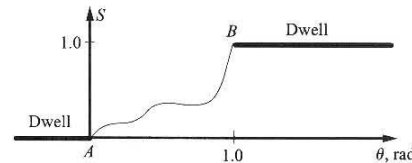
8.8 A cam returns from a full lift of 1.2 in during its initial 60° rotation. The first 0.4 in of the return is half-cycloidal. This is followed by a half-harmonic return. Determine β_1 and β_2 so that the motion has continuous first and second derivatives. Draw a freehand sketch of y' , y'' , and y''' indicating any possible mismatch in the third derivative.



8.9 Assume that s is the cam-follower displacement and θ is the cam rotation. The rise is 1.0 cm after 1.0 rad of rotation, and the rise begins and ends at a dwell. The displacement equation for the follower during the rise period is

$$s = h \sum_{i=0}^n C_i \left(\frac{\theta}{\beta} \right)^i$$

If the position, velocity, and acceleration are continuous at $\theta = 0$, and the position and velocity are continuous at $\theta = 1.0$ rad, determine the value of n required in the equation, and find the coefficients C_i if $\dot{\theta} = 2$ rad/s. Note: Use the minimum possible number of terms.

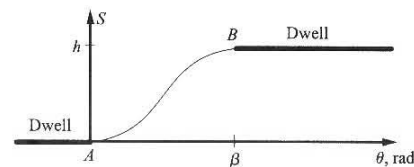


8.10 Re-solve Problem 8.9 if $\theta = 0.8$ rad and $\dot{\theta} = 200$ rad/s.

8.11 For the cam-displacement schedule given, h is the rise, β is the angle through which the rise takes place, and s is the displacement at any given angle θ . The displacement equation for the follower during the rise period is

$$s = h \sum_{i=0}^5 a_i \left(\frac{\theta}{\beta} \right)^i$$

Determine the required values for $a_0, \dots, a_5$ such that the displacement, velocity, and acceleration functions are continuous at the end points of the rise portion.



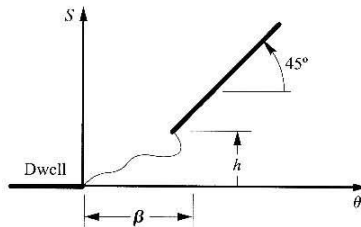
8.12 Re-solve Problem 8.11 if $h = 20$ mm and $\beta = 120^\circ$.

8.13 Re-solve Problem 8.11 if $h = 2$ in and $\beta = 90^\circ$.

8.14 Assume that s is the cam-follower displacement and θ is the cam rotation. The rise is h after β degrees of rotation, and the rise begins at a dwell and ends with a constant velocity segment. The displacement equation for the follower during the rise period is

$$s = h \sum_{i=0}^n C_i \left(\frac{\theta}{\beta} \right)^i$$

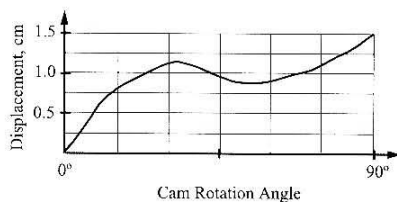
If the position, velocity, and acceleration are continuous at $\theta = 0$ and the position and velocity are continuous at $\theta = \beta$, determine the n required in the equation, and find the coefficients C_i that will satisfy the requirements if $s = h = 1.0$.



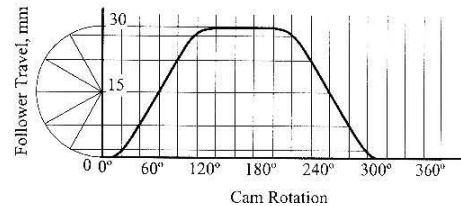
8.15 A follower moves with simple harmonic motion a distance of 20 mm in 45° of cam rotation. The follower then moves 20 mm more with cycloidal motion to complete its rise. The follower then dwells and returns 25 mm with cycloidal motion and then moves the remaining 15 mm with harmonic motion in 45° . Find the intervals of cam rotation for the cycloidal motions and dwell by matching velocities and accelerations, then determine the equations for the displacement (S) as a function of θ for the entire motion cycle.

EXERCISE PROBLEMS ON GRAPHICAL CAM DESIGN

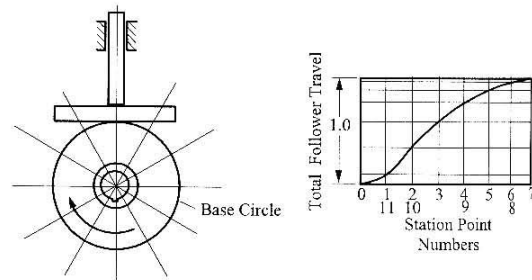
8.16 Construct the part of the profile of a disk cam that follows the displacement diagram shown. The cam has a 5-cm-diameter pitch circle and is rotating counterclockwise. The follower is a knife-edged, radial, translating follower. Use 10° increments for the construction.



8.17 Construct the profile of a disk cam that follows the displacement diagram shown. The follower is a radial roller and has a diameter of 10 mm. The base circle diameter of the cam is to be 40 mm and the cam rotates clockwise.



8.18 Accurately sketch one-half of the cam profile (stations 0–6) for the cam follower, base circle, and displacement diagram given. The base circle diameter is 1.2 in.



8.19 Lay out a cam profile using a harmonic follower displacement (both rise and return). Assume that the cam is to dwell at zero lift for the first 100° of the motion cycle and to dwell at a 1 in lift for cam angles from 160° to 210° . The cam is to have a translating, radial, roller follower with a 1 in roller diameter, and the base circle radius is to be 1.5 in. The cam will rotate clockwise. Lay out the cam profile using 20° plotting intervals.

8.20 Lay out a cam profile using a cycloidal follower displacement (both rise and return) if the cam is to dwell at zero lift for the first 80° of the motion cycle and to dwell at 2 in lift for cam angles from 120° to 190° . The cam is to have a translating, radial, roller follower with a roller diameter of 0.8 in. The cam will rotate counterclockwise, and the base circle diameter is 2 in. Lay out the cam profile using 20° plotting intervals.

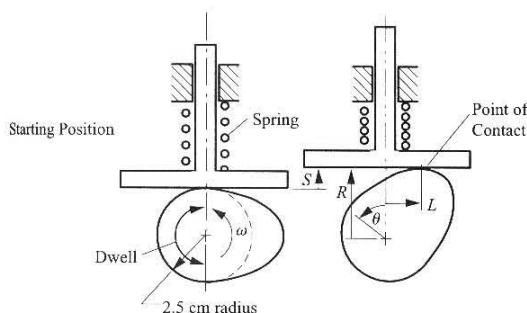
8.21 Lay out a cam profile assuming that an oscillating roller follower starts from a dwell for 0° to 140° of cam rotation, and the cam rotates clockwise. The rise occurs with parabolic motion during the cam rotation from 140° to 220° . The follower then dwells for 40° of cam rotation, and the return occurs with parabolic motion for the cam rotation from 260° to 360° . The amplitude of the follower rotation is 35° , and the follower radius is 1 in. The base circle radius is 2 in, and the distance between the cam axis and follower rotation axis is 4 in. Lay out the cam pro-

file using 20° plotting intervals such that the pressure angle is 0° when the follower is in the bottom dwell position.

8.22 Lay out the rise portion of the cam profile if a flat-faced, translating, radial follower's motion is uniform. The total rise is 1.5 in, and the rise occurs over 100° of cam rotation. The follower dwells for 90° of cam rotation prior to the beginning of the rise and dwells for 80° of cam rotation at the end of the rise. The cam will rotate counterclockwise, and the base circle radius is 3 in.

EXERCISE PROBLEMS ON ANALYTICAL CAM DESIGN

8.23 In the sketch shown, the disk cam is used to position the radial, flat-faced follower in a computing mechanism. The cam profile is to be designed to give a follower displacement S for a CCW cam rotation θ according to the function $S = k\theta^2$ starting from dwell. For 60° of cam rotation from the starting position, the lift of the follower is 1.0 cm. By analytical methods, determine the distances R and L when the cam has been turned 45° from the starting position. Also calculate whether cusps in the cam profile would occur in the total rotation of 60° .



8.24 Determine the cam profile assuming that the translating cylindrical-faced follower starts from a dwell from 0° to 80° , and the cam rotates clockwise. The rise occurs with cycloidal motion during the cam rotation from 80° to 180° . The follower then dwells for 60° of cam rotation, and the return occurs with simple harmonic motion for the cam rotation from 240° to 360° . The amplitude of the follower translation is 3 cm, and the follower radius is 0.75 cm. The base circle radius is 5 cm, and the offset is 0.5 cm.

8.25 Re-solve Problem 8.24 if the amplitude of the follower translation is 4 cm, the follower radius is 1 cm, the base circle radius is 5 cm, and the offset is 1 cm.

8.26 Re-solve Problem 8.24 if the cam rotates counterclockwise.

8.27 Determine the cam profile assuming that the translating flat-faced follower starts from a dwell from 0° to 80° and rotates clockwise. The rise occurs with parabolic motion during the cam rotation from 80° to 180° . The follower then dwells for 60° of

cam rotation, and the return occurs with simple harmonic motion for the cam rotation from 240° to 360° . The amplitude of the follower translation is 3 cm. First find the minimum base circle radius based on avoiding cusps, and use that base circle to design the cam.

8.28 Re-solve Problem 8.27 if the cam rotates counterclockwise.

8.29 Determine the cam profile assuming that an oscillating cylindrical-faced follower dwells while the cam rotates counterclockwise from 0° to 100° . The rise occurs with 3-4-5 polynomial motion during the cam rotation from 100° to 190° . The follower then dwells for 80° of cam rotation, and the return occurs with simple harmonic motion for the cam rotation from 270° to 360° . The amplitude of the follower oscillation is 25° , and the follower radius is 0.75 in. The base circle radius is 2 in, and the distance between pivots is 6 in. The length of the follower is to be determined such that the pressure angle starts out at zero.

8.30 Re-solve Problem 8.29 if the cam rotates clockwise.

8.31 Design a cam and oscillating roller follower assuming that the follower starts from a dwell for 0° to 80° of cam rotation and the cam rotates clockwise. The rise occurs with cycloidal motion during the cam rotation from 80° to 200° . The follower then dwells for 40° of cam rotation, and the return occurs with cycloidal motion for the cam rotation from 240° to 360° . The amplitude of the follower rotation is 45° . Determine the cam base circle radius, distance between cam and follower pivots, the length of the follower, and the radius of the follower for acceptable performance.

8.32 Determine the cam profile assuming that an oscillating flat-faced follower starts from a dwell from 0° to 45° and rotates counterclockwise. The rise occurs with simple harmonic motion during the cam rotation from 45° to 180° . The follower then dwells for 90° of cam rotation, and the return occurs with simple harmonic motion for the cam rotation from 270° to 360° . The amplitude of the follower oscillation is 20° , and the follower offset is 0.5 in. The base circle radius is 5 in, and the distance between pivots is 8 in.

8.33 Re-solve Problem 8.32 if the cam rotates clockwise.

8.34 Design a cam system assuming that an oscillating flat-faced follower starts from a dwell for 0° to 100° of cam rotation and the cam rotates counterclockwise. The rise occurs with uniform motion during the cam rotation from 100° to 200° . The follower then dwells for 40° of cam rotation, and the return occurs with parabolic motion for the cam rotation from 240° to 360° . The oscillation angle is 20° .

8.35 Design a cam system assuming that an oscillating flat-faced follower starts from a dwell for 0° to 50° of cam rotation and the cam rotates clockwise. The rise occurs with cycloidal motion during the cam rotation from 50° to 200° . The follower then dwells for 90° of cam rotation, and the return occurs with harmonic motion for the cam rotation from 290° to 360° . The oscillation angle is 25° .

8.36 Determine the cam profile assuming that the translating knife-edged follower starts from a dwell from 0° to 80° and rotates clockwise. The rise occurs with cycloidal motion during the cam rotation from 80° to 180° . The follower then dwells for 60° of cam rotation, and the return occurs with simple harmonic motion for the cam rotation from 240° to 360° . The amplitude of the follower translation is 4 cm. The base circle radius is 5 cm, and the offset is 0.5 cm.

8.37 A radial flat-faced follower is to move through a total displacement of 20 mm with harmonic motion while the cam rotates through 30° . Find the minimum radius of the base circle that is necessary to avoid cusps.

8.38 A radial flat-faced follower is to move through a total displacement of 3 in with cycloidal motion while the cam rotates through 90° . Find the minimum radius of the base circle that is necessary to avoid cusps.

8.39 A radial roller follower is to move through a total displacement of $L = 19$ mm with harmonic motion while the cam rotates 60° . The roller radius is 5 mm. Use the program supplied with the book and find the minimum radius necessary to avoid cusps during the interval indicated.

8.40 A radial roller follower is to move through a total displacement of $L = 45$ mm with cycloidal motion. The roller radius is 5 mm, and the cam rotates 90° during the rise. Use the program supplied with the book and find the minimum radius necessary to avoid cusps during the interval.

8.41 Assume that a flat-faced translating follower is used with the displacement schedule in Problem 8.10. Determine if a cusp is present at $\theta = 60^\circ$.

8.42 Assume that a flat-faced translating follower is used with the displacement schedule in Problem 8.12. Determine if a cusp is present at $\theta = 90^\circ$.